

**DESIGN AND IMPLEMENTATION OF A MACHINE
LEARNING-POWERED SURVEILLANCE SYSTEM FOR
PREDICTING AND VISUALIZING INFECTIOUS DISEASES
OUTBREAKS: MALARIA AND CHOLERA IN KENYA**

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*A research project submitted to the department of Information Technology in
the School of Computing and Information Technology in partial fulfillment of
the requirement for the award of a Bachelor's Degree in Information
Technology of Jomo Kenyatta University of Agriculture and Technology*

DECLARATION

I hereby declare that this research project is my original work and has not been presented for a degree in any other University.

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This project has been submitted for examination with my approval as University Supervisor.

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ABSTRACT

Infectious diseases, specifically malaria and cholera, remain significant public health challenges in Kenya, with 75% of the population at risk of malaria and recurrent cholera outbreaks intensifying in underserved areas (World Health Organization, 2024). Despite the digital transformation under the Digital Health Act (2023), infectious disease management remains trapped in a reactive cycle. Current surveillance systems, such as DHIS2 and HMIS, rely on "lagging indicators"—recording cases only after patients present at clinics—meaning that by the time data is aggregated and analyzed, outbreaks are often already in an exponential growth phase. There is a critical research gap in developing an integrated, automated framework that seamlessly merges real-time climate data with routine health records to provide simultaneous early warning signals for both diseases across all 47 counties in Kenya.

This research proposes a full-stack machine learning-powered surveillance system designed to transition disease monitoring from reactive reporting to proactive prediction. The system utilizes a unified platform to synthesize historical health records, climatic data (rainfall, temperature, humidity), and population density to generate actionable "risk scores". By integrating Long Short-Term Memory (LSTM) neural networks for time-series forecasting and Geographic Information Systems (GIS) for geospatial visualization, the platform will enable health authorities to identify and visualize transmission "hotspots" in real-time.

The project adopts CRISP-DM (Cross-Industry Standard Process for Data Mining) system development approach, following iterative phases of business understanding, data understanding, data preparation, modeling, evaluation and deployment.

The expected outcome is a web-based predictive analytics platform that provides interactive dashboards for public health officials. The system will deliver validated, county-specific forecasts, allowing for the proactive allocation of medical resources such as vaccines and antimalarials before an outbreak reaches its peak. Ultimately, this research aims to provide a scalable, intelligent health system that reduces morbidity and mortality by enhancing the speed and accuracy of infectious disease detection in Kenya.

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Acronyms

Acronym	Definition
ADASYN	Adaptive Synthetic Sampling
ANN	Artificial Neural Network
ARIMA	Adaptive Reference Machine Learning
AUC	Area Under the Curve
CHAP	Climate Health Analytics Platform
CNN	Convolutional Neural Network
CORP	Cholera Outbreak Risk Prediction
DHIS2	District Health Information Software 2
DL	Deep Learning
DT	Decision Trees
EHR	Electronic Health Record
EPSG	European Petroleum Survey Group (coordinate system reference)
HIE	Health Information Exchange
HMIS	Health Management Information System
IDSR	Integrated Disease Surveillance and Response
KeNADA	Kenya National Data Archive
KHIS	Kenya Health Information System
K-NN	K-Nearest Neighbors
LR	Logistic Regression
MLP	Multilayer Perceptron
MSE	Mean Squared Error
NCDC	Nigerian Centre for Disease Control
NiMet	Nigerian Meteorological Agency
PCA	Principal Component Analysis
PEM	Predictive Epidemic Modeling
RFR	Random Forest Regression
RLR	Regularized Linear Regression
RNN	Recurrent Neural Network
SMOTE	Synthetic Minority Oversampling Technique

SVM	Support Vector Machine
WASH	Water, Sanitation, and Hygiene
XGBoost	Extreme Gradient Boosting

Definition of Terms

Algorithm: A mathematical procedure used to identify patterns in data during the machine learning process.

Artificial Intelligence (AI): A branch of computer science that enables machines to simulate human intelligence, including learning, reasoning, and decision-making, used in this study for disease outbreak prediction.

Class Imbalance: A condition in epidemiological datasets where negative cases (no outbreak) significantly outnumber positive cases (outbreak), which can bias machine learning model predictions.

Climatic Drivers: Physical climatic conditions such as temperature, rainfall, and relative humidity that influence the spread and intensity of malaria and cholera outbreaks.

Compartmental Model: A mathematical model that divides a population into distinct groups based on disease status, such as Susceptible, Infected, and Recovered (SIR), to understand disease transmission dynamics.

Data Preprocessing: The process of cleaning, normalizing, formatting, and integrating raw data into a unified pipeline to prepare it for machine learning model training.

Deep Learning (DL): A specialized sub-field of machine learning that uses multi-layered artificial neural networks to solve complex, non-linear problems such as disease outbreak prediction and medical imaging.

Disease Surveillance: The continuous and systematic collection, analysis, and interpretation of health data used to monitor and respond to disease trends and outbreaks in a population.

Early Warning System: A proactive mechanism that detects and signals the likelihood of a disease outbreak before it reaches its peak, enabling timely public health intervention.

Endemic Zone: A geographic area where a disease such as malaria is constantly present or regularly found among the population.

Feature: An individual measurable input variable or predictor used by a machine learning model to make predictions, such as rainfall, temperature, or population density.

Full-Stack System: A complete software system that encompasses both the frontend user interface and the backend server, database, and machine learning components working together as a unified platform.

Geographic Information System (GIS): A computer-based system used to capture, store, analyze, and visualize spatial or geographic data, used in this study to map and display disease outbreak hotspots across Kenyan counties.

Geospatial Data: Data that contains geographic or location-based information, used in this study to map disease distributions and identify high-risk transmission areas.

Hotspot: A geographic area identified as having a significantly higher risk or concentration of disease transmission compared to surrounding areas.

Label (Target Variable): The specific outcome a machine learning model is trained to predict, in this case the risk level of a malaria or cholera outbreak classified as either high risk or low risk.

Lagging Indicator: A measure that records events only after they have already occurred, such as disease cases recorded only after patients present at health facilities, limiting early outbreak detection.

Long Short-Term Memory (LSTM): A type of recurrent neural network architecture specifically designed to learn patterns in sequential and time-series data, used in this study to forecast disease outbreak trends over time.

Machine Learning (ML): A sub-field of Artificial Intelligence that enables algorithms to learn directly from data and make predictions or decisions without being explicitly programmed for specific tasks.

Morbidity: The incidence or prevalence of a disease or medical condition within a population, used in this study as a measure of the burden of malaria and cholera in Kenya.

Mortality: The number of deaths caused by a disease within a population over a given period, used as an indicator of the severity of malaria and cholera outbreaks in Kenya.

Outbreak: A sudden increase in the number of cases of a disease above what is normally expected in a specific area or population within a given time period.

Overfitting: A condition where a machine learning model learns the specific patterns and noise of the training data too closely, resulting in poor performance when applied to new, unseen data.

Predictive Analytics: The use of statistical algorithms, machine learning models, and historical data to forecast future events or trends, in this case the likelihood of disease outbreaks.

Public Health Informatics: An interdisciplinary field that applies information science, computer science, and public health principles to improve the collection, management, and use of health data for decision-making.

Real-Time Data: Data that is collected, processed, and made available for use immediately or with minimal delay, enabling timely monitoring and response to disease outbreaks.

Risk Score: A numerical value generated by the predictive model that indicates the likelihood of a disease outbreak occurring in a specific geographic area based on environmental and health data inputs.

Supervised Machine Learning: A type of machine learning where a model is trained using labelled historical datasets containing known input variables and expected outputs, to learn patterns and make predictions on new data.

Time-Series Forecasting: A technique used to predict future values based on previously observed data points collected over time, applied in this study to forecast disease outbreak trends using historical health and climate data.

Unsupervised Machine Learning: A type of machine learning where algorithms analyzed and group unlabelled data based on inherent patterns or similarities, without relying on predefined outcomes or target labels.

Vector: In the context of infectious diseases, a living organism such as the Anopheles mosquito that transmits a pathogen such as the malaria parasite from one host to another.

Web-Based Platform: A software application or system that is accessed and operated through a web browser over the internet, used in this study to deliver the predictive analytics dashboards to public health officials.

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Infectious diseases are illnesses caused by pathogenic microorganisms such as bacteria, viruses, parasites or fungi and can be transmitted from one person to the other either directly or indirectly or through environmental factors and with potential for global spread and impact (World Health Organization, 2023). Despite the advancements in medical science, infectious diseases such as malaria and cholera still pose a significant burden on global health systems and landscape, majorly in low and middle-income countries. According to the World Health Organization (WHO), infectious diseases are among the leading causes of morbidity and mortality globally, with malaria and cholera being notable examples. Research has shown that the major factors contributing to the increasing prevalence of these diseases include climate change, globalization and environmental factors (Siiba et al., 2024). Globally, health organizations have adopted disease surveillance systems and digital platforms to monitor, detect, and respond to outbreaks of infectious diseases (World Health Organization, 2022). However, many of these systems and platforms remain to be largely reactive, relying on historical data and delayed reporting, which limits their effectiveness in early disease outbreak, detection and prevention.

In Kenya, infectious diseases like malaria and cholera pose significant public health challenges, heavily impacting vulnerable populations. The Ministry of Health (MoH) in Kenya plays a crucial role in managing these diseases through surveillance, prevention, containment and treatment strategies. Despite these efforts, persistent gaps in the public health systems hinder effective prediction and outbreak management of such infectious diseases. The Kenya Ministry of Health relies on the Integrated Disease Surveillance and Response (IDSR) framework, supported by the Health Management Information System (HMIS) and the District Health Information Software (DHIS2), primarily relying on health facility reporting and community-based surveillance. However, these systems often suffer from delays, underreporting, and lack of real-time data, hampering effective outbreak management.

By 2026, the focus of Kenyan public health system has shifted toward Predictive Analytics. The introduction of the DHIS2, Climate Health Analytics Platform (CHAP) and collaborative

efforts between the Kenya Meteorological Department and KEMRI (Kenya Medical Research Institute) mark a transition toward using Machine Learning (ML) to forecast disease risks based on climate variables (e.g., rainfall anomalies and temperature deviations). However, there remains a critical research gap in developing an integrated, automated framework that seamlessly merges real-time climate data with routine health facility records to provide actionable early warning signals for both Malaria and Cholera simultaneously (Bukhair & Ajisafe, 2025).

1.2 Project Overview

This study is situated within the multidisciplinary field of Public Health Informatics and Data Science, specifically focusing on the development of Predictive Epidemic Modeling (PEM). It aims to bridge the gap between traditional epidemiology and modern computational intelligence by creating a full-stack analytics framework tailored for infectious disease surveillance.

Globally, the landscape of infectious disease surveillance is undergoing a paradigm shift. In 2026, the artificial intelligence (AI) market for disease outbreak prediction is estimated to exceed \$3 billion, driven by the increasing frequency of climate-induced health crises. International health bodies, including the WHO Hub for Pandemic and Epidemic Intelligence, are advocating for the integration of "Big Data"—encompassing environmental sensors, mobility patterns, and real-time electronic health records (EHRs) to move beyond reactive reporting. Advanced nations have begun implementing cloud-native, AI-driven platforms like BlueDot and HealthMap, which scan global data to identify emerging threats weeks before traditional clinical detection.

In Kenya, the post-COVID-19 era and the enactment of the Digital Health Act (2023) have laid the foundation for a national Health Information Exchange (HIE). Despite this, regional health surveillance remains primarily descriptive. For instance, while the Kenya Meteorological Department and KEMRI have launched early warning systems for the Western Highlands, these models are often siloed and lack a unified "full-stack" interface for county-level health officers. Malaria and Cholera continue to be exacerbated by erratic rainfall patterns and urban density, highlighting a critical need for a localized, integrated platform that translates complex data into actionable public health "risk scores" across all 47 counties.

The proposed system aims at utilizing a full-stack architecture to integrate data from various sources such as historical health data, climatic and environmental factors. The proposed system will deploy Machine Learning & Time-Series Forecasting models such as Long-Short Term Memory to identify patterns between environmental anomalies (temperature/rainfall) and disease spikes. It aims to integrate Geospatial metadata that is Geographical Information Systems coordinate system (EPSG:4326) for mapping county boundaries and provide visualization of real-time risk maps. The proposed system will enable a proactive intervention strategy, ensuring that medical resources are deployed before an outbreak reaches its peak.

1.3 Statement of the problem

Infectious diseases continue to pose a significant global public health challenge despite advancements in medical research and surveillance systems. According to the World Health Organization, there were approximately 282 million cases of Malaria and about 610,000 related deaths worldwide in 2024, with over 90% of the burden (265 million) occurring in Africa (WHO, 2025). Similarly, Cholera remains a major global public health concern, with more than 560,000 cases and over 6000 deaths reported in 2024, particularly in regions with inadequate water and sanitation systems (WHO, 2024). These statistics demonstrate the persistent and widespread impact of infectious diseases globally.

In Kenya, the burden of infectious diseases remains high, with 75% of the population at risk of the disease. Malaria is endemic in many parts of the country, with an estimated 3.3 million cases and over 1,000 deaths reported annually (WHO, 2024). Cholera outbreaks also occur periodically, especially in densely populated and underserved areas, contributing to increased morbidity and strain on healthcare systems. The country reported approximately 8,926 cases and 145 deaths in 2023, with outbreaks intensifying in 2022-2023, exceeding 12,000 cases and over 200 deaths (Kiama et al., 2025). Despite efforts by the Ministry of Health and county governments to strengthen disease surveillance, the existing systems are largely reactive, relying on historical data and delayed reporting mechanisms.

Despite the digital transformation of Kenya's healthcare sector under the Digital Health Act (2023), infectious disease management remains trapped in a reactive cycle. Additionally, the reliance on reactive surveillance systems such as the DHIS2 and HMIS results in delays in outbreak detection, limited real-time insights, and ineffective early warning capabilities. Studies have shown that surveillance systems in many developing countries suffer from incomplete and untimely reporting, which significantly hinders timely response to disease

outbreaks (Malebana et al., 2025). Consequently, outbreaks are often identified after they have already spread, leading to increased transmission rates, higher healthcare costs, and preventable loss of lives.

The magnitude of this problem is evident in the frequent recurrence of outbreaks, overstretched healthcare facilities, and reduced effectiveness of intervention strategies. Currently, surveillance systems rely on "lagging indicators"—cases are only recorded after patients present at health facilities. By the time this data is aggregated and analyzed at the county level, an outbreak is often already in its exponential growth phase. Furthermore, current surveillance approaches do not adequately leverage modern technologies such as machine learning and data analytics to predict outbreaks or identify high-risk areas. This limits the ability of health authorities to make proactive, data-driven decisions toward public health response.

Therefore, the core research problem lies in the absence of an adequately integrated, full-stack machine-learning predictive analytics platform that leverages Long Short-Term Memory (LSTM) networks and Geospatial metadata (EPSG:4326) to synthesize historical health records with real-time environmental variables, capable of predicting and visualizing malaria and cholera outbreaks in Kenya. Addressing this gap is essential to enhance early detection, improve response time, and reduce the overall impact of infectious diseases, particularly malaria and cholera in Kenya.

1.4 Proposed Solution

To address this challenge, this study proposes a machine learning-powered surveillance system designed to predict and visualize infectious disease outbreaks, specifically for malaria and cholera across Kenyan counties, leveraging Long Short-Term Memory (LSTM) neural networks and geospatial mapping. This system will integrate multiple data sources, including climatic data, population density, and historical outbreak records, to develop predictive models to forecast potential outbreaks, providing real-time risk assessments to aid public health decision-making. The proposed system will also provide interactive dashboards for health officials to visualize outbreaks and allocate resources efficiently, ultimately improving public health outcomes.

1.5 Objectives

1.5.1 General Objective

The main aim of this research proposal is to develop a full-stack machine learning-powered surveillance system for predicting and visualizing the outbreak of malaria and cholera across Kenyan counties.

1.5.2 Specific Objectives

1. To collect and preprocess multi-source data, including historical health data, environmental, and geospatial data, to identify key factors influencing outbreaks of Malaria and Cholera in Kenya.
2. To design a full-stack system architecture and a machine learning-based predictive model for forecasting the risk of malaria and cholera outbreaks.
3. To develop a web-based predictive analytics system for visualizing disease outbreak risks at the county level in Kenya.
4. To evaluate the performance of the predictive model using historical datasets and appropriate metrics such as prediction accuracy and model validation techniques.

1.6 Research Questions

1. How does historical health data, environmental, and geospatial data influence the outbreak of Malaria and Cholera in Kenya?
2. How can a full-stack system architecture and machine learning models be optimized to process multi-dimensional geospatial and health data for reliable outbreak forecasting?
3. In what ways can a web-based visualization framework improve the interpretability of complex predictive analytics for public health decision-makers at the county level?
4. What is the predictive reliability and accuracy of the developed machine learning models when validated against historical outbreak data using standard statistical error metrics?

1.7 Justification

Infectious diseases continue to pose a significant burden to global health systems and within Kenya as well. This underscores the urgent need for the adoption of more effective predictive and proactive infectious diseases surveillance systems. Infectious diseases such as Malaria and Cholera still record significant morbidity and mortality rates, especially in low and medium-income countries with limited healthcare infrastructure and poor sanitation. This is

despite the existence of surveillance systems which rely on "lagging indicators" (reporting cases only after they appear in clinics), retrospective data and delayed reporting mechanisms, limiting their ability to effectively provide predictive early warnings to support timely response and interventions.

This research is justified by the need to address the limitations posed by the existing Malaria and Cholera surveillance systems which are insufficient in the prediction and visualization of the infectious diseases. By leveraging emerging technologies such as machine learning and data analytics, the proposed system aims to transform disease surveillance from a reactive to a predictive approach. This will enable early detection of potential outbreaks, allowing health authorities to implement preventive measures before widespread transmission occurs.

The proposed solution will directly benefit key stakeholders, including the Ministry of Health, county health departments, vulnerable populations, and public health practitioners, by providing a tool that enhances decision-making through real-time risk prediction and visualization. Additionally, communities will benefit from improved outbreak preparedness and reduced exposure to infectious diseases. The system also contributes to more efficient allocation of healthcare resources by identifying high-risk areas in advance.

From a research perspective, this study contributes to the growing field of intelligent health systems by integrating machine learning techniques with spatial data analysis for disease prediction. While previous studies have explored disease surveillance, there is limited implementation of integrated predictive systems within the Kenyan context. Therefore, this research fills an important gap by demonstrating how advanced technologies can be applied to improve public health outcomes in developing countries.

Furthermore, the relevance of this study is underscored by current global trends emphasizing the use of digital technologies and artificial intelligence in healthcare. As the world continues to face emerging and re-emerging infectious diseases, there is a critical need for innovative, data-driven and predictive approaches to disease monitoring and control. This study is therefore timely and aligns with ongoing efforts to strengthen health systems through technological innovation.

1.8 Proposed System Methodologies

The proposed methodology for the development of this system will follow the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework, which is widely used in data science and machine learning projects. This methodology is suitable for this project because it provides a structured, iterative approach specifically tailored for data-driven systems such as the prediction and visualization of infectious disease outbreaks. CRISP-DM emphasizes continuous refinement, understanding of data, and alignment between technical processes and real-world objectives, making it ideal for developing a predictive analytics platform for Malaria and Cholera outbreaks in Kenya. The development will take place in the following phases iteratively:

1.8.1 Business Understanding

This phase will involve clearly defining the objectives of the system in relation to public health needs. The goal of the project is to develop a machine learning-powered surveillance system capable of predicting and visualizing Malaria and Cholera outbreaks across Kenya. During this phase, key stakeholders such as public health officials will be identified, and their requirements will be gathered to ensure the system addresses real-world challenges. The expected outcomes, including early outbreak detection, improved decision-making, and effective resource allocation, will also be established.

1.8.2 Data Understanding

In this phase, relevant data sources will be identified and collected to support the prediction of disease outbreaks. This will include historical disease records, weather data, population density statistics, and environmental factors that influence the spread of Malaria and Cholera. The collected data will be explored to understand its structure, quality, and relevance to the project objectives. Initial data analysis will be performed to identify patterns, trends, and potential issues such as missing values or imbalanced datasets.

1.8.3 Data Preparation

This phase will involve transforming the raw data into a clean and usable format suitable for machine learning. Data preprocessing tasks such as cleaning, normalization, integration, and formatting will be carried out to ensure consistency and accuracy. Tools such as Python, Pandas, and NumPy will be used to manipulate and process the datasets efficiently. Feature engineering techniques will also be applied to extract meaningful variables that improve the predictive performance of the model.

1.8.4 Modeling

During the modeling phase, machine learning techniques will be applied to develop predictive models for disease outbreaks. The project will utilize LSTM (Long Short-Term Memory) neural networks implemented using TensorFlow or Keras to handle time-series forecasting. Different model configurations and parameters will be tested and optimized to achieve the best performance. The model will be trained using the prepared dataset and evaluated iteratively to improve accuracy.

1.8.5 Evaluation

In this phase, the developed model will be thoroughly evaluated to ensure it meets the project objectives and performs effectively in predicting disease outbreaks. Performance metrics such as accuracy, precision, recall, and F1-score will be used to assess the model. In addition to technical evaluation, the system will also undergo user validation through feedback from public health officials to ensure usability and relevance. Any necessary improvements will be identified before deployment.

1.8.6 Deployment

This will be the final phase which will involve deploying the system as a full-stack web-based platform accessible to users. The system will be hosted on cloud platforms such as Heroku or Render to ensure scalability and availability. The deployment will include the integration of the machine learning model, backend APIs, database, and frontend dashboard with geospatial visualization tools like Leaflet or Mapbox. Continuous monitoring and maintenance will be carried out to update datasets, improve model performance, and ensure the system remains effective over time. Version control and collaboration will be managed using Git and GitHub to support future enhancements.

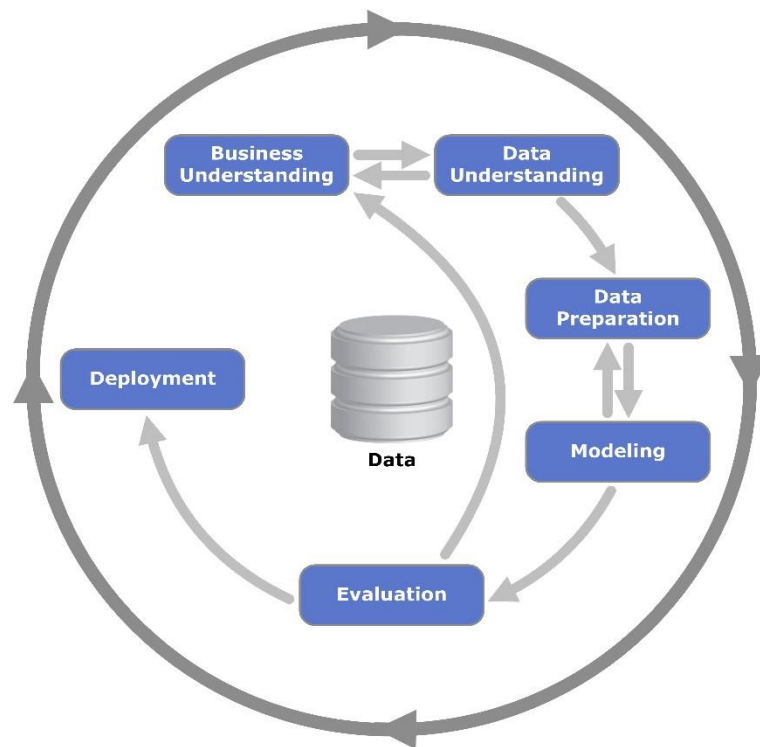


Figure 1-1: CRISP-DM Methodology

1.9 Scope

This study focuses on the development of a machine learning-powered surveillance system to predict the outbreak of infectious diseases, specifically malaria and cholera across the Kenyan Counties. The research will make use of the available historical health datasets including, and real-time data related to climatic factors such as rainfall, temperature and humidity as well as environmental factors such as water sources and sanitation conditions.

The target population for this study are the communities living in the high-risk areas for malaria and cholera prevalence as well as public health stakeholders whom the surveillance system will help gain predictive insights and visualization of disease trends thereby helping in proactive response to the outbreaks.

The project will be limited to the development of a machine learning-surveillance system with integrated Geographical Information System visualization component for mapping the distributions of disease outbreaks and identifying potential hotspots. The study will not encompass the diagnosis of diseases, disease treatment or the pharmaceutical interventions for malaria and cholera.

Furthermore, this study acknowledges the following potential limitations in relation to the field of study:

1. Resource Constraints: This study acknowledges computational resource constraints and the constraints in the optimization of the machine learning model.
2. Environmental Determinism: the models that will be used will primarily focus on climatic and environmental drivers. As a result, socio-economic factors occurrences like the sudden changes in migrations or local sanitation conditions among communities will not be captured I the predictive model.
3. Methodological Constraint: the machine learning model will identify the relations between climatic and environmental factors and the spread of malaria and cholera, but will not prove the biological causes of these diseases.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter aims to present a comprehensive understanding and review of the existing knowledge and research related to the adoption and application of machine learning, data-driven systems and geospatial technologies in the prediction and visualization of malaria and cholera outbreaks. The review evaluates how emerging technologies such as artificial intelligence, Geographical Information Systems and predictive analytics are being adopted in disease surveillance and decision-making to enhance public health response in the management of the outbreak of these diseases.

This study focuses on the adoption and application of machine learning techniques and models in the prediction of infectious diseases, specifically malaria and cholera in Kenya. These diseases have been monitored and their outbreaks managed largely using traditional surveillance systems which have posed challenges such as being slow, reactive and significant reporting lags. The integration and application of machine learning techniques and models addresses the systematic vulnerabilities of traditional surveillance systems through its capability to analyze vast, heterogeneous datasets including satellite-based climate data socio-economic indicators, and electronic health records—to identify complex patterns and anomalies indicative of an outbreak (Tshimula et al., 2024). Machine learning and Artificial Intelligence integration into the surveillance of infectious diseases highlights a remarkable shift in the prediction and management of the outbreaks of malaria and cholera particularly in regions with limited resources like the sub-Saharan Africa (Tshimula et al., 2024).

There has been a growing adoption and application of machine learning models, geospatial and real-time predictive analytics to improve the forecasting and visualization of the outbreaks of malaria and cholera in Kenya. Machine learning (ML) and Artificial Intelligence (AI) are particularly valuable in this setting and are becoming rapidly applied in healthcare since these technologies can analyze vast data collections to uncover insights that are hidden which are missing in traditional statistical methods (Muriithi et al., 2024). Machine Learning models provide the ability to draw knowledge from data and identify relevant patterns using classification methods. The application of Machine Learning algorithms can be of value to healthcare providers in the identification of effective measures and in the decision-making process for the best public health response (Nkiruka et al., 2021).

This literature review aims to evaluate the existing studies and systems that are related to the prediction and visualization of the outbreak of infectious diseases such as malaria and cholera. The review also serves a purpose in identifying and assessing the methodologies and technologies that have been used in the application of machine learning to predict these outbreaks, as well as identifying the gaps that the proposed system seeks to address. Particularly, this literature review seeks to:

1. Examine the existing surveillance systems and approaches to the prediction of malaria and cholera outbreak.
2. Assess the application of machine learning, artificial intelligence and GIS visualization in disease prediction and surveillance.
3. Identify the limitations and shortcomings of the existing prediction and surveillance systems.
4. Justify the need for a more integrated, rapid data-driven framework, real-time, and geospatially enabled predictive platform for the prediction of malaria and cholera outbreak in Kenya.

2.2 Theoretical Review

This section presents the core concepts forming the foundation of Machine Learning applied in the development of a prediction and visualization system of malaria and cholera. It also assesses the key theoretical divisions defining the application of machine learning in the prediction of infectious diseases. Additionally, it highlights the advantages and shortcomings of each kind of the theoretical divisions defining this field, particularly in the prediction and analytics of the outbreaks of infectious diseases.

2.2.1 Key Concepts and Variables

Several core and foundational machine learning concepts are defined as follows to provide an understanding of the design and development of a machine learning-powered surveillance system for malaria and cholera. The key environmental and socio-economic variables that are critical in the prediction of the outbreaks of malaria and cholera are as well defined.

1. Machine Learning (ML)

Machine Learning is the sub-field of Artificial Intelligence that deals with the construction of algorithms that are capable of learning directly from data without

relying on other predetermined equations (Leo, 2020). An algorithm is the mathematical procedure used to identify patterns in data.

1. Data

Data refers to raw facts and figures collected from various sources such as historical health records, climatic data, and population statistics. In this study, data includes malaria and cholera case records, rainfall levels, temperature, humidity, population statistics, and sanitation indicators.

2. Features (Predictors)

Features are the input parameters or individual independent variables used by a model to make a prediction (Sadiq et al., 2024). In the Kenyan context, key features include climatic factors such as rainfall, temperature, humidity, geographic data such as endemic zones, altitude, and socio-economic indicators such as access to basic sanitation, population density.

3. Labels (Target Variable)

The label is the specific outcome the model is designed to predict (Muriithi et al., 2024). The label in this case is the risk level of disease outbreak which is either high risk or low risk for malaria and cholera outbreak.

4. Training and Testing

Training involves feeding historical data into a model so its algorithm can learn patterns, while testing evaluates how well the model performs on new, unseen data.

5. Performance Metrics

Performance metrics are quantitative measures of model performance that are used to measure the effectiveness of a surveillance system. The metrics that will be used include:

- i. **Accuracy:** Refers to the overall proportion of correct predictions that are made by the machine learning model (Muriithi et al., 2024).
- ii. **Sensitivity:** This refers to the ability of the model to identify true positive cases correctly, which is important in the medical/healthcare context to avoid missing actual infections (Leo, 2020).
- iii. **Specificity:** This is the ability of the prediction model to identify individuals who are disease-free that is the true negatives (Muriithi et al., 2024).

- iv. F1-Score: The weighted harmonic mean of precision and recall, providing a balanced measure for models tested on imbalanced datasets (where one class significantly outnumbers the other) (Muriithi et al., 2024)

6. Geographic Information Systems (GIS)

GIS refers to computer systems used to capture, store, analyze, and visualize spatial or geographic data. In this study, GIS is used to map and visualize disease outbreaks across different counties in Kenya.

7. Climatic Drivers

These are the physical climatic conditions that form primary determinants in influencing the spread of malaria and cholera in the context of the prediction of these diseases. They include temperature, precipitation (rainfall), and relative humidity.

8. Weather Drivers

These are the climatic factors that influence the occurrence, spread, and intensity of events or conditions especially in the outbreaks of diseases. For example, higher temperatures and excessive rainfall favor the growth of *Vibrio cholerae* which is the bacterium responsible for the infectious disease cholera.

2.2.2 Theoretical Frameworks for Infectious Disease Prediction

2.2.2.1 Statistical and Epidemiological Models

Statistical/Epidemiological Models are defined as the logical and mathematical demonstrations of diseases epidemiology and their related processes (Çağırğan & Çağırğan, 2020). These models also rely on statistical relationships to describe disease dynamics and identify key factors influencing outbreaks of diseases such as malaria and cholera. The models focus on identifying and ranking the mechanisms that are responsible for disease transmission and provide the necessary tools for the interpretation of studies related to epidemiology (Jewell, 2003). This school of thought relies on formalising the relationships between variables through fixed mathematical equations (Inam, 2025). These models are widely used in epidemiology for analysis, forecasting, and public health planning and decision-making. While traditional models such as mathematical, statistical and epidemiological models have heavily relied on differential equations and compartmental structures, recent studies highlight a paradigm shift toward incorporating advanced machine learning (ML) techniques to enhance the predictive accuracy (Inam, 2025).

The Statistical/Epidemiological Models are further classified into several broad categories based on their objectives and structures as follows:

- i. **Compartmental Models:** Compartmental models are a subset of mathematical models that divide/partition a population into distinct groups known as compartments based on disease status. A common framework used in the compartmental models is the SIR framework which is expressed as:



Figure 2-1: SIR Framework

Where:

S (Susceptible): Individuals who can contract the disease.

I (Infected): Individuals who have the disease and can transmit it.

R (Recovered): Individuals who have recovered and may have immunity.

Disease transmission occurs through interactions between compartments, and individuals transition from one state to another based on defined rates. Compartmental models are used to understand the spread of diseases like cholera (through contaminated water) and malaria (through vector transmission), helping to estimate outbreak magnitude and duration.

- ii. **Deterministic and Stochastic Models:** Deterministic models assume that disease predictions follow fixed mathematical relations between different elements and provide the same result for the same starting conditions meaning that the same input will always produce the same output (Çağırğan & Çağırğan, 2020). Stochastic models include elements that are random and have chances, which makes them more realistic since they reflect the variability of nature, also providing a range of possible outcomes (Çağırğan & Çağırğan, 2020).
- iii. **Static and Dynamic Models:** Static models represent the condition of a system at a given point in time, while dynamic models account for changes that have taken place over a specific interval. Dynamic modelling is preferred for scenarios like

vaccination, where incidence varies over time before settling (Çağırğan & Çağırğan, 2020).

- iv. **Network Models:** These model view individuals as nodes and their interactions as edges in a graph. Network models represent the spread of diseases as a process that occurs through connections or relationships (edges) between individuals or locations (nodes). Instead of assuming everyone mixes equally, this approach models who interacts with whom, making it more realistic for transmission dynamics (Craig et al., 2020).

Advantages of Statistical and Epidemiological Models

- i. **Predictive Power and Decision Support:** the models enable public health officials to assess the behavior beforehand using presumptive scenarios, therefore assisting in the development of vaccination, treatment, and public health strategies (Çağırğan & Çağırğan, 2020).
- ii. **Uncertainty Elimination:** The models provide ways of overcoming uncertainties by combining experimental data, reviews by experts on the dynamics of infections and field of knowledge to predict uncertainties in the occurrence of infectious diseases.
- iii. **Interventions Evaluation:** These models are crucial in the testing of various options of control (e.g., patient isolation, face mask policies, the closure of schools) to identify which measure are most effective at decreasing the spread of infectious diseases.
- iv. **Analytical Clarity:** The models enable the microscopic explanation of macroscopic disease behaviour and help measure the uncertainties inherent in future predictions (Çağırğan & Çağırğan, 2020).

Limitations of Statistical and Epidemiological Models

- i. **Data Quality Issues:** These models are heavily constrained by data quality and availability. Issues like underreporting, misdiagnosis, and reporting delays (where data trickles in over several days) can lead to spurious conclusions.
- ii. **Usability Barriers:** Traditional mathematical models are often complex to understand and extend for non-mathematicians, requiring deep knowledge of model equations.

- iii. **Assumption Sensitivity:** Simple models (like the basic SIR) often make unrealistic assumptions, such as homogeneous mixing or immediately starting infectiousness, which can dwarf noise-related uncertainty.
- iv. **Dependent Happening:** Unlike non-communicable diseases (e.g., lung cancer), observation of one infected individual is not independent of others in communicable diseases, which confuses the assessment of population effectiveness.

2.2.2.2 Supervised Machine Learning

Recent studies have identified Supervised Machine Learning as being the most efficient prediction paradigm for predicting climate-sensitive diseases as well as infectious diseases, particularly in resource-limited countries like Sub-Saharan Africa (Inam, 2025). Supervised machine learning is defined by the use of labeled datasets containing historical input data such as climatic or socio-economic variables and target labels (such as "positive" or "negative" test results) to guide the algorithm's learning process (Ileperuma et al., 2023). It involves training a model using labeled historical data, where the input variables (features) and the expected output (label) are already known. The model is trained using historical datasets and then tested to evaluate its prediction accuracy. By analyzing past data, the model identifies patterns and relationships that can be used to make predictions on new, unseen data (Hu et al., 2025). In disease prediction, this means using historical data (e.g., climate, demographics, past cases) to predict whether or when diseases like malaria or cholera will occur.

Recent studies demonstrate that supervised ML consistently outperforms traditional statistical and mathematical models in providing quick and accurate disease forecasts (Tshimula et al., 2024). The following recent studies have explained the performance of Supervised Machine learning in the prediction of diseases:

1. **Malaria Prediction:** A 2024 study in Kenya found that the Random Forest algorithm was the most robust for predicting malaria occurrence, achieving a classification accuracy of 97.33% (Muriithi et al., 2024). In Senegal, Random Forest models reached accuracy levels above 0.9 in regions with high prevalence by analyzing satellite-derived rainfall data (Ileperuma et al., 2023). Additionally, ML has been used to predict mosquito bed net utilization with 83% accuracy (Tariq et al., 2025).
2. **Cholera Outbreak Forecasting:** The Cholera Outbreak Risk Prediction (CORP) model, utilizing supervised techniques, achieved a remarkable 99.62% accuracy in

Nigeria (Tshimula et al., 2024). Other 2024 research in Nigeria using Random Forest achieved 92% accuracy and integrated user-friendly interfaces for real-time alerts (Omankwu & Etuk, 2024).

3. **COVID-19 Tracking:** In Zambia, the XGBoost algorithm was identified as the best performer for predicting mortality in hospitalised patients, reaching 92.3% accuracy (Tshimula et al., 2024). In South Africa, Artificial Neural Networks (ANN) and Random Forest were used to identify advanced age and severe symptoms as top predictors of ICU mortality (Tshimula et al., 2024).

Advantages of Supervised Machine Learning

- i. **Accuracy and Robustness:** Models learn directly from historical labeled data, which improves reliability. Tree-based ensemble models, such as Random Forest and XGBoost, are noted for being comparatively insensitive to covariation among predictors and for delivering superior results on performance measures like AUC-ROC (Inam, 2025).
- ii. **Early Warning and Proactive Response:** Can predict outbreaks before they happen using past trends. ML models facilitate real-time analytics, allowing health authorities to allocate resources (such as vaccines and antimalarials) to "hotspots" before an outbreak peaks (Inam, 2025).
- iii. **Data Integration:** These algorithms can process vast, heterogeneous datasets, including satellite imagery, genomic data, and electronic health records, to uncover hidden patterns that miss in traditional methods (Inam, 2025).
- iv. **Handling Complexity:** Unlike traditional models (e.g., ARIMA or SIR), supervised ML can capture complex, non-linear interactions between climate variables and disease incidence (Inam, 2025).

Limitations of Supervised Machine Learning

Despite its potential, several barriers hinder the widespread adoption of supervised learning in low-resource settings:

- i. **The "Black Box" Problem:** High-performing models, especially deep learning and complex ensembles, often lack interpretability. This lack of transparency can erode

trust among policymakers who must make critical decisions based on model outputs (Obeagu & Ogenyi, 2025).

- ii. **Data Quality and Dependency:** These systems rely on large volumes of high-quality labeled data (Inam, 2025). In many African contexts, health information systems are fragmented, and underreporting can lead to biased model conclusions (Obeagu & Ogenyi, 2025).
- iii. **Class Imbalance:** Most epidemiological datasets contain a vast majority of negative cases (e.g., 96% negative). Without advanced sampling techniques like SMOTE or ADASYN, supervised models tend to be biased toward the majority class, failing to predict actual outbreaks (Leo, 2020).
- iv. **Overfitting:** There is a constant risk that a model will learn the specific "noise" of the training data rather than generalisable patterns, leading to poor performance on new datasets (Sadiq et al., 2024).

2.2.2.3 Unsupervised Learning

The main principle of unsupervised learning is that the algorithm analyzes input data and automatically groups or organizes it based on similarities or patterns without prior knowledge of outcomes. Since 2022, the application of unsupervised machine learning (ML) in disease surveillance has been recognised as a vital tool for exploratory data analysis, particularly when dealing with the high-dimensional and often unlabelled datasets common in low-resource settings (Ileperuma et al., 2023). Unsupervised learning differs from supervised methods by seeking out hidden patterns, structures, or groupings within data without relying on target labels. In disease outbreak prediction, recent studies have shown that its role expands beyond data exploration:

1. **Hotspot Identification and Clustering:** Hotspot Identification and Clustering: Algorithms like K-means are increasingly used to cluster regions by infection rates, providing insights that support effective resource allocation and targeted interventions (Inam, 2025). For instance, in Kenya, recent efforts have used these techniques to map cholera hotspots by integrating epidemiological persistence with Water, Sanitation, and Hygiene (WASH) indicators (Wamae et al., 2026).
2. **Dimensionality Reduction:** Techniques such as Principal Component Analysis (PCA) are employed to handle "excessive features" in complex datasets, identifying

the most critical factors influencing disease incidence—such as climatic and socio-economic variables—before supervised models are applied (Inam, 2025).

3. **Medical Imaging:** Recent research highlights the use of unsupervised feature learning for cystoscopic image classification and the unsupervised segmentation of 3D medical images based on clustering and deep representation learning (Intorne et al., 2023).
4. **Diarrheal Disease Determinants:** A 2023 review noted the use of **cluster analysis** to determine the socio-economic and environmental drivers of diarrheal diseases in resource-poor settings.
5. **Vector Surveillance:** In Iran (2025), a two-stage monitoring approach used Random Forest and MLP (often informed by unsupervised feature extraction) to identify high-risk depressions for mosquito breeding habitats, achieving an AUC score of 94%.

Advantages of Unsupervised Machine Learning

- i. **Pattern Discovery:** Capable of discovering patterns and groupings in datasets that lack labels, which is essential for novel or emerging zoonoses.
- ii. **Exploratory Power:** Vital for identifying meaningful trends that inform the derivation of new research questions and future analyses.
- iii. **Resource Strategy:** Effective at clustering geographical areas by risk level, allowing health officials to station staff proactively during climate emergencies.

Limitations of Unsupervised Machine Learning

- i. **Interpretability:** These models often lack interpretability, making it difficult for public health decision-makers to understand the rationale behind a specific cluster
- ii. **Validation:** It is significantly harder to validate results compared to supervised learning because there is no "ground truth" or historical label to compare against.
- iii. **Accuracy:** For direct outbreak prediction (Yes/No), unsupervised methods generally underperform compared to supervised models.

2.2.2.4 Deep Learning (Architectural Intelligence)

Deep learning (DL) is a specialized sub-field of Artificial Intelligence (AI) and Machine Learning (ML) that utilizes multi-layered artificial neural networks to solve complex, non-linear problems (Sadiq et al., 2024). It is inspired by biological neural structures in the human brain, where layers of interconnected "neurons" process information in stages (Stewart, 2022). In this framework, each layer of the network learns to extract higher-order features

from the previous layer, allowing the system to automatically recognize patterns without being explicitly programmed for specific features (Ljubic et al., 2022).

Deep learning has outperformed traditional machine learning in several domains due to its ability to handle multimodal and high-dimensional data (Stewart, 2022). It employs complex architectures like:

- i. **Convolutional Neural Networks (CNNs):** These are specifically designed for spatial analysis and are the gold standard for medical imaging. For example, CNN architectures like MobileNetV2 and ResNet50 have achieved up to 97.06% accuracy in detecting malaria parasites from microscopic blood smear images.
- ii. **Recurrent Neural Networks (RNNs):** These excel at capturing temporal sequences and time-dependent data. Units like Long Short-Term Memory (LSTM) are frequently used for forecasting disease outbreaks (such as COVID-19 or influenza) by analyzing how contagion unfolds over time.
- iii. **Transformers:** Emerging architectures that excel at integrating diverse datasets (e.g., social, demographic, and environmental) to provide a complete view of disease spread.

Advantages of Deep Learning

- i. **Higher Accuracy:** DL models can capture complex, non-linear relationships that traditional models often oversimplify.
- ii. **Feature Learning:** They reduce the need for manual feature extraction by learning feature hierarchies directly from the data.

Limitations of Deep Learning

- i. **"Black Box" Nature:** A significant limitation is the lack of interpretability, making it difficult for clinicians to understand the rationale behind a prediction.
- ii. **Resource Intensity:** These models involve high computational costs and require specialized hardware (like GPUs) and massive datasets for effective training.
- iii. **Data Requirements:** DL typically requires large-scale labeled datasets; for smaller or sparse datasets, traditional ML methods remain more feasible.

2.3 Case Study Review

This section examines the adoption of existing applications of machine learning, predictive analytics and geospatial systems in the surveillance of infectious diseases, specifically focusing on malaria and cholera outbreaks prediction and visualization. The review of the case studies highlights some of the real-world implementations, successes that were achieved and their limitations in relation to the proposed machine learning surveillance system for malaria and cholera prediction and visualization in Kenya. The following case studies have been reviewed in relation to the application of machine learning in the prediction and visualization of malaria and cholera outbreaks in Kenya.

1. Machine Learning for Malaria Prediction in Kenya

This case study, based on research by Muriithi et al. (2024), evaluates the implementation of supervised machine learning to predict malaria occurrence—specifically final test results—within the Kenyan context. The study is significant because it aims to complement existing public health efforts by providing a robust, data-driven approach to mitigate malaria-related threats, which remain a major concern in Kenya’s tropical and endemic zones.

The researchers utilized a well-documented dataset from the Kenya National Data Archive (KeNADA) and investigated twelve predictor variables. The research and implementation involved estimating and comparing the performance of the following five distinct supervised machine learning algorithms:

1. Support Vector Machines (SVM)
2. K-Nearest Neighbors (K-NN)
3. Random Forest
4. Tree Bagging
5. Boosting

The study adopted a binary classification approach (0=Negative, 1=Positive) using repeated cross-validation (five times) to ensure the models were validated against varied data subsets. The core discussion of the results is centered on the ability of these models to analyze both linear and non-linear relationships between features, which allows for deeper insights into complex epidemiological patterns than traditional statistical methods.

Key Successes

The primary success of the study and implementation was the identification of Random Forest as the superior and most robust model for malaria classification. Key performance milestones included:

- i. **High Accuracy:** The model attained a classification accuracy of 97.33%.
- ii. **Robust Metrics:** The model demonstrated a specificity of 98.4% and an Area Under the Curve (AUC) of 98.3%, indicating a high ability to distinguish between infected and non-infected individuals.
- iii. **Critical Feature Identification:** The model successfully ranked the importance of predictors, revealing that the presence of *Plasmodium falciparum* was the most critical feature (100% relative importance), followed by geographic region, endemic zone, and anaemic levels.
- iv. **Resource Optimization:** By identifying high-risk regions and biological predictors, the system enables health authorities to proactively allocate resources to areas with the highest susceptibility.

Possible Misses and Limitations

While the study and implementation achieved high accuracy, it highlighted several "misses" or complexities that reflect the challenges of machine learning in healthcare:

- i. **Unexpected Feature Irrelevance:** A notable finding was that having a mosquito bed net was found to have 0% relative importance in predicting final test results within this specific model, a result that might seem counterintuitive to traditional prevention strategies.
- ii. **Model Sensitivity Discrepancy:** While accuracy was high, the sensitivity (recall) was lower at 71.1%, suggesting the model was slightly less effective at correctly identifying all positive cases compared to its high specificity for negative cases.
- iii. **Inconsistency with Previous Literature:** The findings contradicted some earlier studies; for instance, while this research favored Random Forest, previous literature had found that Naïve Bayes outperformed other models in predicting outbreaks.

- iv. **Dataset Constraints:** The study's conclusions are tied to the specific characteristics of the KeNADA dataset, and the performance of such models must be continuously monitored as new data is incorporated to maintain predictive relevance over time.

2. Machine Learning for Cholera Outbreak Prediction in Nigeria

This case study, based on the research by Omankwu and Enefiok (2024), evaluates the development and implementation of a data-driven framework for the early detection and prediction of cholera outbreaks in Nigeria. The study is particularly relevant given Nigeria's diverse climatic landscape—ranging from arid northern regions to humid tropical southern areas—which complicates the transmission dynamics of cholera.

The researchers developed a predictive model by integrating diverse, multimodal datasets to capture the complex drivers of cholera with the following aspects of implementation:

- i. **Data Integration:** The system pulled historical outbreak records from the Nigerian Centre for Disease Control (NCDC), weather data (daily rainfall and temperature) from the Nigerian Meteorological Agency (NiMet), and socio-economic indicators like population density and sanitation infrastructure from the National Bureau of Statistics and local agencies.
- ii. **Methodology:** The implementation followed a systematic pipeline of data preprocessing (handling missing values via mean substitution and regression imputation), feature engineering (creating interaction terms between rainfall and temperature), and model training.
- iii. **Algorithm Comparison:** The study experimented with several supervised learning algorithms, including Support Vector Machines (SVM), Neural Networks, and Random Forest.

The core discussion highlights that while traditional statistical models often lack the precision required for timely interventions, Machine Learning algorithms are uniquely suited for this task due to their ability to identify non-linear patterns across climatic and demographic variables.

Key Successes

The primary success of the implementation was the superior performance of the Random Forest algorithm, which proved to be the most robust for the Nigerian context with the following:

- i. **High Performance Metrics:** The Random Forest model achieved an accuracy of 92%, a precision of 90%, a recall of 88%, and an F1-score of 89%.
- ii. **Identification of Key Predictors:** The model successfully identified rainfall, temperature, and population density as the most influential features for predicting outbreaks.
- iii. **Practical Deployment (User Interface):** A significant milestone was the creation of a user-friendly interface for public health officials. This platform provides real-time alerts and geospatial visualizations through GIS technology, allowing authorities to monitor hotspots and allocate resources like medical supplies and vaccines more effectively.
- iv. **Validation:** The model's predictions showed a strong correlation with actual historical outbreak data, validating its accuracy and reliability.

Possible Misses and Limitations

Despite the technical successes realized in this study, the study identified several critical limitations and challenges that impact the system's reliability:

- i. **Data Quality Dependency:** The model's predictive accuracy is highly contingent on the quality and completeness of the input data. Issues with missing information and data inconsistencies in regional records remain a bottleneck.
- ii. **Data Integration Complexity:** Combining datasets from multiple fragmented government agencies and international organisations requires complex data-sharing mechanisms that are not yet fully standardised.
- iii. **Missing Variables:** The current model excludes potentially vital data sources, such as high-resolution satellite imagery or social media sentiment analysis, which could further enhance early warning signals.

- iv. **Need for Continuous Updates:** To remain relevant, the model requires constant re-training with new data to adapt to evolving environmental conditions and changes in human behaviour or sanitation infrastructure

3. Machine Learning for Predicting Residual Malaria in Jazan, Saudi Arabia

This case study which was based on research by Sadiq et al. (2024), explores the application of machine learning (ML) to predict residual malaria infections in the Jazan region of Saudi Arabia. This research is particularly significant as it focuses on a low malaria-transmission region, aiming to use environmental factors to bolster eradication efforts.

The researchers implemented a comparative framework to determine which model best captures the relationship between malaria incidence and climate with the following key implementations:

- i. **Data Integration:** The study utilized a dataset of malaria infections confirmed via microscopic examination of blood smears (the gold standard) from 2015 to 2020. These records were combined with numerical meteorological data, including average temperature, maximum temperature, and relative humidity.
- ii. **Methodology:** The dataset was split into an 80:20 ratio for training and testing/validation. The implementation involved three distinct ML models: Artificial Neural Networks (ANN), Random Forest Regression (RFR), and Regularized Linear Regression (RLR).
- iii. **Algorithm Performance:** The findings suggested that the ANN model was the most effective for this task. This is attributed to the ANN's superior ability to capture complex, non-linear relationships inherent in climate-disease data, which traditional linear models or even some tree-based models might miss

Key Successes

- i. **Superior Accuracy:** The ANN provided better predictions than other methods, evidenced by the lowest error across evaluation metrics such as Mean Squared Error (MSE).
- ii. **Early Warning Potential:** The study successfully demonstrated that monitoring environmental variables in real-time can allow authorities to respond swiftly to emerging threats, potentially preventing epidemics before they occur.

- iii. **Optimized Resource Allocation:** By identifying regional risk factors, the model provides public health authorities with a data-driven basis for directing resources (e.g., medical supplies and vector control) to where they are most needed.
- iv. **Capturing Complexity:** The implementation highlighted the strength of AI in dealing with the complex biological and environmental interactions that define malaria transmission in a specific region.

Possible Misses and Limitations

Despite the technical successes, the study identified several "misses" and barriers to broader application:

- i. **Region-Specific Constraints:** The findings are highly region-specific to Jazan; the model may not perform accurately if applied to areas with different environmental or ecological conditions without further localized training.
- ii. **Data Quality Issues:** The model's accuracy was limited by the quality of data sourced from external databases, which suffered from inconsistencies or a lack of localized factors.
- iii. **Excluded Variables:** A significant "miss" was the focus solely on meteorological factors. The researchers noted that the model excluded other critical variables, such as socio-economic conditions, human mobility, and specific vector control measures already in place.
- iv. **Technical Risks (Overfitting):** While the ANN was flexible and accurate, the study warned that such models are prone to overfitting if not tuned with extreme care, which can compromise their reliability on new, unseen data.
- v. **Static Forecasts:** The long-term forecasts generated by the system may fail to account for future shifts in healthcare infrastructure or changes in national medical interventions

4. Malaria outbreak prediction in The Gambia

This case study, based on the research by Khan et al. (2024), evaluates the implementation of machine learning (ML) techniques to predict malaria outbreaks across every district in The Gambia. The study is a pioneering effort in the region, aiming to provide health authorities with an early warning system to mitigate the high seasonal malaria prevalence, which typically peaks during and immediately after the rainy season.

The researchers developed a predictive framework by integrating two distinct longitudinal datasets spanning from 2013 to 2021. The model utilized historical meteorological data (rainfall, relative humidity, and maximum/minimum temperatures) and clinical data (monthly malaria cases and deaths) sourced from the Gambian Health Management Information System. In this research, a key aspect of the implementation was the inclusion of non-climatic features, specifically district-level population size and the month of the year, which were found to be significant in predicting when epidemics would occur.

The study compared the performance of eight supervised learning algorithms: C5.0 Decision Trees (DT), Artificial Neural Networks (ANN), K-Nearest Neighbors (KNN), Logistic Regression (LR), Extreme Gradient Boosting (XGBoost), Random Forest (RF), and Support Vector Machines (SVM) with both linear and radial kernels. To ensure reliability and prevent overfitting, the models were trained using repeated 10-fold cross-validation (repeated five times).

Key Successes

The implementation demonstrated that machine learning can provide highly accurate district-specific forecasts for malaria outbreaks as follows:

- i. **Top Performers:** XGBoost and C5.0 Decision Trees emerged as the most effective models for the Gambian context.
- ii. **High Performance Metrics:** On the testing dataset, both top models achieved a classification accuracy of 93.3% and a sensitivity (recall) of 96.7%. High sensitivity is particularly valuable in public health as it ensures that very few potential outbreaks are missed.
- iii. **Discriminatory Power:** XGBoost demonstrated the highest capability to distinguish between outbreak and non-outbreak periods, with an Area Under the Curve (AUC) of 0.97.
- iv. **Pattern Recognition:** The models successfully captured the complex, non-linear interplay between environmental variables, identifying that the highest malaria incidence consistently occurs between August and December.

Possible Misses and Limitations

Despite the technical successes, the study identified several "misses" and systemic challenges:

- i. **Data Imbalance:** A major hurdle was the significant imbalance in the dataset, where "no outbreak" periods far outnumbered "outbreak" events, a common issue in epidemiological modeling that can bias results.
- ii. **Excluded Biological/Intervention Factors:** The current study did not address other critical predictors such as the use of insecticide-treated bed nets, the coverage of indoor residual spraying, or the emergence of drug resistance.
- iii. **Generalization Challenges:** The findings are region-specific; because climatic impacts vary significantly by geography, the models may require extensive localized retraining before they can be applied to different countries.
- iv. **Methodological Inconsistencies:** The researchers noted that differences in datasets and reporting methods across the literature make it difficult to compare these results directly with other global malaria prediction studies

2.4 Integration and Architecture

This section outlines how the proposed machine learning-powered surveillance system for the prediction and visualization of malaria and cholera can be integrated with existing sources of data and implemented with the appropriate architectures of system implementation. The architectural frameworks of system architectures that are robust and support scalability, multimodal datasets, processing in real-time, and effective visualization necessary for public health response and decision-making are also explored. Architectural frameworks and integration strategies that are essential for connecting advanced machine learning predictive models with the existing public health systems and infrastructure are highlighted as well.

The integration and architecture of a machine learning-powered surveillance system for malaria and cholera in Kenya rely on multi-layered frameworks that bridge data collection, processing, and real-time user interaction (Kiama et al., 2023). Implementing these systems involves several technical options and architectural designs to ensure scalability, security, and interoperability with existing public health infrastructure.

2.4.1 System Integration Approaches

The proposed machine learning-powered surveillance system requires proper integration of multimodal data sources and services in order to achieve accurate prediction and visualization of the outbreaks of malaria and cholera in Kenya. Traditional surveillance systems for the prediction and visualization of infectious diseases pose challenges such as slow reporting times, limited predictive capability and no real-time integration of climate + health data (Maddah et al., 2023). The limitations posed by traditional infectious diseases surveillance systems such as Kenya Health Information System (KHIS) and DHIS2 underscore the need for more advanced predictive models that can be integrated with the existing public health infrastructure and multimodal datasets to improve response times in the management of malaria and cholera outbreaks in Kenya.

Integrating complex machine learning (ML) models into functional public health applications requires seamless data flow between various tools and platforms.

2.4.1.1 Integration with Existing Health Information Systems

The machine learning-powered surveillance system can be integrated with existing platforms such as the Kenya Health Information System (KHIS) to obtain historical disease records and the District Health Information System (DHIS) to provide predictive monitoring and tracking of outbreaks. This allows automatic or semi-automated data exchange between hospitals, county health departments, and the predictive system.

2.4.1.2 Integration with Weather and Climate APIs

Application Programming Interfaces (APIs) are essential for allowing different systems to share data and interoperate (Muriithi et al., 2024). Real-time environmental data such as rainfall, temperature, and humidity can be integrated using external APIs such as OpenWeatherMap. These data inputs are critical for predicting malaria and cholera outbreaks since both diseases are strongly influenced by climatic conditions.

2.4.1.3 Integration with GIS Platforms

Integrating GIS technology allows for the visualization of spatial distributions and the mapping of transmission "hotspots" to guide targeted resource allocation (Anthony, 2023). The system will integrate Geographic Information System (GIS) tools such as Leaflet.js or Google Maps API to visualize disease distribution across Kenyan counties. This enables spatial representation of outbreak hotspots for decision-making.

2.4.1.4 API-Driven Connectivity

A RESTful API architecture will be used to connect the frontend, backend, machine learning model, and external data sources. For instance, ML models developed in Python using the Flask framework can be connected to user-facing interfaces built with libraries like React. This approach allows developers to integrate AI technology regardless of their specific expertise in machine learning. This ensures seamless communication between system components and supports real-time data exchange.

2.4.2 System Architecture Options

There are different design architectures that can be adopted in the implementation of the proposed system. A robust and effective framework for disease outbreak prediction typically follows a structured, modular design to allow for flexibility and rapid development and often include specific core functions such as detection, reporting, confirmation, and feedback mechanisms. They must also address "cross-border" collaboration and multisectoral coordination to be effective in large-scale public health scenarios. The most suitable system architecture options are microservices architecture and the layered architecture.

2.4.2.1 Layered Architecture (3-Tier Architecture)

The layered architecture is a common approach for full-stack systems such as the proposed machine learning-powered surveillance system. This architecture consists of three functional layers which work in coordination to deliver the system functionalities as follows:

- i. **Presentation Layer (Frontend):** This layer consists of the frontend including the user interface for map visualization, analytics of predictions and dashboards.
- ii. **Application Layer (Backend):** This is the system layer that handles the business logic of the system, processing of the data and communication with the predictive machine learning model.
- iii. **Data Layer (Database):** This system architecture layer consists of the database which stores data such as the historical diseases data, environmental data and the results from the prediction.

The layered system architecture in the implementation of the proposed system will be suitable since it is simple to design and implement, easy to maintain and debug errors as well as having a clear separation of concerns in the implementation of the system. However, this

architecture has limitations such as complexity issues when integrating many services crucial for the system implementation.

2.4.2.2 Microservices Architecture

In this system architecture, the proposed system can be implemented into independent divisions which represents the independent system services such as the prediction service, data ingestion service and the visualization service. Each system service communicates with the use of APIs. The microservices system architecture is highly scalable and flexible even for large systems, easier to update individual system components and suitable for real-time and large-scale systems . On the flip side, the microservices system architecture poses complexity in the design and management, and requires strong API management and deployment strategies.

2.4.2.3 Hybrid Architecture

The proposed machine learning-powered surveillance system can adopt a hybrid architecture combining layered system architecture and the microservices approaches. This approach ensures both scalability and simplicity in the implementation of the system. An example of a hybrid system architecture can as follow this approach:

- i. Core system (frontend, backend, database) uses layered architecture
- ii. Machine learning and external integrations operate as microservices

2.5 Summary

The literature review reveals a significant shift in infectious disease surveillance, transitioning from traditional reactive systems to proactive, predictive data-driven frameworks powered by machine learning (ML) and artificial intelligence (AI).

The literature identifies a paradigm shift where traditional models are being supplemented or replaced by advanced computational techniques. The review demonstrate that Statistical and Epidemiological models are historically the foundational models that excel at describing the biological processes but struggle with non-linear climate-disease interaction, also lacking real-time integration capabilities.

Studies suggest that supervised machine learning is the most efficient paradigm for climate-sensitive diseases in regions with limited resources. Supervised machine learning consistently outperforms traditional models through capturing complex patterns in multimodal datasets.

Deep Learning (Architectural Intelligence) utilizes architectures like CNNs for spatial analysis and LSTMs for temporal sequences, thereby achieving diagnostic accuracies of up to 97.06% in malaria parasite detection.

Hybrid and Network Models are emerging schools of thought focusing on fusing mechanistic biological models with AI to maintain interpretability while enhancing predictive power.

Recent studies and implementations relevant to the application of machine learning in the prediction of infectious diseases demonstrate the practical success and limitations of these technologies across Africa as highlighted below:

1. In Kenya, a study about machine learning in the prediction of malaria and cholera by Muriithi et al. (2024) identified Random Forest machine learning algorithm as the most robust model for classifying test results, achieving 97.33% accuracy. Random Forest algorithm ranked the presence of *Plasmodium falciparum* and geographic region as critical predictors.
2. In Nigeria, research by Omankwu and Enefiok (2024) about machine learning in the prediction of cholera outbreaks developed a framework using Random Forest that achieved 92% accuracy. A major success was the integration of a user-friendly GIS interface for real-time alerts.
3. In Tanzania, the Adaptive Reference Machine Learning (ARML) model leveraged XGBoost to achieve 78.5% accuracy. It utilized ADASYN to handle class imbalance and PCA for dimensionality reduction.
4. In Gambia, the first study of its kind in the region used XGBoost and C5.0 Decision Trees to achieve 93.3% accuracy in predicting district-level outbreaks, emphasizing the importance of both climatic and non-climatic features like population size.

2.6 Research gaps

1. Geographical Granularity and Lack of Actionable Visualization

In Kenya, regional health surveillance remains primarily descriptive, often lacking the granularity needed for county-level decision-making. Surveillance systems are frequently disconnected, leading to limited information at the sub-county level and delays in reporting.

This research seeks to address this by developing a web-based predictive analytics system designed to visualize disease trends across all 47 Kenyan counties. By integrating Geographic Information Systems (GIS) and interactive dashboards, the

system provides real-time geospatial visualizations of "hotspots" to aid health officials in efficient resource allocation.

2. **Absence of Simultaneous Multi-Disease Surveillance**

There is a critical research gap in developing automated frameworks capable of providing early warning signals for both malaria and cholera simultaneously within a single platform.

The general objective of this research is to develop a unified machine learning-powered surveillance system specifically tailored for predicting and visualizing outbreaks of both diseases across the Kenyan landscape.

3. **Technical Barriers (Data Imbalance and Sparsity)**

Epidemiological modeling in developing countries is severely hindered by "broken" datasets characterized by missing information, data inconsistency, and extreme class imbalances—where negative cases can represent up to 96% of the data. Traditional models fail to perform adequately under these conditions, often leading to biased conclusions.

The research proposal addresses this through a rigorous Data Preprocessing and Reengineering phase. It employs advanced sampling and balancing techniques—such as ADASYN or SMOTE—to restore sampling balance and handle dataset complexities before model training begins.

4. **Reactive Systems and Dependency on "Lagging Indicators"**

Traditional surveillance systems in Kenya, such as DHIS2 and HMIS, are primarily reactive, meaning they rely on historical data and "lagging indicators" where cases are only recorded after patients present at clinics. By the time this data is aggregated and analyzed, outbreaks are often already in an exponential growth phase, leading to preventable loss of life and overstretched facilities.

This research seeks to resolve this by implementing a proactive intervention strategy. It utilizes Long Short-Term Memory (LSTM) neural networks to identify patterns between real-time environmental anomalies (such as temperature and rainfall) and subsequent disease spikes, allowing medical resources to be deployed before an outbreak peak.

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This chapter presents the system analysis and design of the proposed machine learning-powered surveillance system for predicting and visualizing malaria and cholera outbreaks in Kenya. It begins by describing the systems development methodology adopted for this research, followed by a comprehensive feasibility study assessing the viability of the proposed system from economic, technical, and operational perspectives. The chapter further details the requirements elicitation process used to identify both functional and non-functional system requirements, outlines the proposed system architecture, and presents the physical design of the system including the database schema and user interface wireframes. Together, these components provide a structured blueprint that guides the design and implementation of the full-stack predictive analytics platform.

3.2 System Development Methodology

This research adopts the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework as the systems development methodology. CRISP-DM is a widely used, vendor-neutral process model specifically designed for data science and machine learning projects. Unlike traditional software development methodologies, CRISP-DM is structured around the data and the analytical problem rather than software delivery cycles, making it particularly appropriate for this research project centered on building a predictive machine learning model.

The methodology is iterative in nature, meaning that phases can be revisited and refined as new insights emerge during development. This iterative design is especially valuable in this project, where model performance outcomes during the Evaluation phase may necessitate revisiting earlier phases such as Data Preparation or Modeling to improve predictive accuracy.

CRISP-DM organizes the development process into six phases, which will be followed throughout this research as follows:

- **Business Understanding:** Defining the project objectives in terms of public health needs, identifying key stakeholders such as county health officials, and establishing expected system outcomes including early outbreak detection and resource allocation support.

- **Data Understanding:** Identifying and collecting relevant data sources including historical disease records from DHIS2/KHIS, climatic data from the Kenya Meteorological Department, population density statistics, and environmental indicators. Initial exploration will assess data quality, completeness, and relevance.
- **Data Preparation:** This phase will involve transforming the raw data into a clean and usable format suitable for machine learning. Data preprocessing tasks such as cleaning, normalization, integration, and formatting will be carried out to ensure consistency and accuracy.
- **Modeling:** Developing and training the LSTM neural network models for time-series forecasting of malaria and cholera outbreak risk, implemented using TensorFlow/Keras, with iterative tuning of hyperparameters to optimize performance.
- **Evaluation:** Assessing model performance using metrics such as accuracy, precision, recall and F1-score, alongside user validation with public health officials to confirm system usability and real-world relevance.
- **Deployment:** Deploying the full-stack platform on, integrating the machine learning model, FastAPI backend, SQLite database, and Dash/Plotly dashboard with geospatial visualization capabilities.

The CRISP-DM framework is well-suited to this research because it provides a structured yet flexible process explicitly tailored for data-driven systems, ensures continuous alignment between technical development and real-world public health objectives, and supports the iterative refinement that is inherent in machine learning model development.

3.3 Feasibility Study

Before proceeding with the design and implementation of the proposed system, a feasibility study was conducted to evaluate the practicality and viability of the project. This analysis examines whether the system can be successfully built and deployed within the constraints of available resources, technology, and time, and whether it will deliver meaningful value to its intended users. The findings of each dimension of the feasibility study are presented below:

3.3.1 Economic Feasibility

The proposed system has been designed with cost-efficiency as a guiding constraint, leveraging open-source tools and freely available datasets throughout the technology stack. There are no licensing costs associated with Python, TensorFlow, Keras, FastAPI, SQLite, or Leaflet.js. Cloud hosting and compute resources required for model training and system

deployment can be sourced from free-tier platforms such as Google Colab, Render, or Railway, keeping operational costs minimal.

As outlined in the project budget (Appendix II), the estimated total cost of the project is Ksh. 76,000, covering hardware (a development laptop at Ksh. 60,000), internet connectivity for three months (Ksh. 6,000), printing and documentation (Ksh. 2,000), and research and miscellaneous costs (Ksh. 8,000). Software costs are zero given the open-source stack.

From a benefits perspective, the economic case for this system is compelling. By enabling early detection of malaria and cholera outbreaks, the system supports proactive allocation of medical resources such as vaccines, antimalarials, and personnel before outbreaks peak. This has the potential to significantly reduce the healthcare costs associated with managing full-scale outbreaks, as well as the indirect economic costs of disease burden such as lost productivity and increased mortality in high-risk communities. The system is therefore economically feasible and offers a strong return on investment relative to its development cost.

3.3.2 Technical Feasibility

The proposed system relies on technologies and tools that are well-established, openly available, and widely used in production-grade data science and web development environments. Python, together with libraries such as TensorFlow, Keras, Pandas, and NumPy, provides a mature ecosystem for building and training the LSTM predictive models at the core of the system. FastAPI offers a high-performance, lightweight framework for exposing the machine learning model as a RESTful API, while Flask and Plotly provide robust support for interactive web-based dashboards and geospatial visualizations.

The data required for the system that is historical disease records, climatic variables, and geospatial county boundaries is accessible through established sources including DHIS2/KHIS, the Kenya Meteorological Department, and public satellite data repositories. Integration with these sources via APIs and direct data exports is technically achievable with the selected stack.

The primary technical challenge lies in the availability of high-quality, consistently formatted historical data and the computational requirements for training LSTM models. These challenges are manageable: data preprocessing pipelines using Pandas will address quality and formatting issues. On the basis of the above, the project is technically feasible.

3.3.3 Operational Feasibility

The proposed system is designed to serve public health officials at the county and national levels within Kenya's Ministry of Health. Its primary interface which is an interactive web-based dashboard with geospatial maps and risk score visualizations is intentionally designed for accessibility and interpretability by non-technical users. Public health officers will not require expertise in machine learning or data science to operate the system; they need only interpret the visual outputs and act on the risk indicators provided.

Training and onboarding requirements are therefore minimal. Moreover, the system does not seek to replace existing workflows but to augment them by adding a predictive layer on top of the existing reporting infrastructure.

Operationally, the system will require periodic maintenance to retrain the predictive models as new data becomes available and to update data pipelines in response to changes in source APIs or data formats. This is manageable within the scope of the research and can be documented as part of the deployment handover. On the basis of the above assessment, the system is operationally feasible.

3.3.4 Schedule Feasibility

Schedule feasibility evaluates whether the proposed system can be developed and implemented within the available project timeline. It assesses the practicality of completing all planned activities, including system analysis, design, development, testing, and documentation, within the allocated duration.

The project timeline, as captured in the Gantt Chart (Appendix III), allocates sufficient time across the key phases of development: data collection and preprocessing, model development and training, backend and frontend development, integration and testing, and final documentation. The iterative nature of the CRISP-DM methodology further supports schedule feasibility by allowing phases to be revisited and refined concurrently rather than following a rigid sequential path, reducing the risk of timeline overruns caused by unforeseen challenges in model performance or data quality. The project is therefore feasible within the proposed schedule.

3.4 Requirements Elicitation

Requirements elicitation was carried out to gather information necessary for understanding the needs of the system's intended users and to identify the functional and non-functional requirements that would guide the design and implementation of the machine learning-

powered surveillance system for predicting and visualizing the outbreaks of malaria and cholera. The process was guided by the research objectives and aimed at understanding user needs, technical constraints, and real-world expectations for a machine learning-powered infectious disease surveillance and outbreak prediction platform in the Kenyan public health context. The process involved engaging key stakeholders — primarily public health officials, epidemiologists, and health data officers — who interact directly with existing disease surveillance infrastructure such as DHIS2 and HMIS and secondary data analysis.

3.4.1 Data Collection Instruments

Two data collection instruments were employed: structured questionnaires and secondary data analysis.

3.4.1.1 Structured Questionnaires

A structured questionnaire was developed and administered to a broader group of respondents including county health information officers, public health nurses, and disease surveillance coordinators. The questionnaire employed a five-point Likert scale to capture respondents' perceptions of the adequacy of existing surveillance systems, their comfort with digital health platforms, and the importance they assigned to specific system features such as real-time risk maps, county-level forecasts, and outbreak alert notifications. Closed-ended questions were used to ensure consistency and ease of quantitative analysis. The questionnaire was administered both physically and through a Google Form shared via official county health department communication channels. The questionnaire instrument is attached as Appendix IV.

3.4.1.2 Secondary Data Analysis

Secondary data analysis was conducted to complement the primary data gathered through questionnaires, and to inform the data requirements of the machine learning predictive models. Existing datasets and documented sources were reviewed, including historical malaria and cholera case records from the Kenya Health Information System (KHIS) and DHIS2, climatic data from the Kenya Meteorological Department, population density statistics from the Kenya National Bureau of Statistics, and published surveillance reports from the Ministry of Health and WHO. This analysis provided insight into the structure, quality, and availability of the datasets that the system would rely upon, and helped identify gaps such as missing values, inconsistent reporting periods, and class imbalance in outbreak

records — all of which directly informed the data preprocessing and feature engineering requirements of the system.

3.4.2 Sampling Technique and Sample Size Determination

The target population for this study comprised public health professionals within Kenya's county health departments and national-level institutions, including the Ministry of Health and KEMRI, who are involved in disease surveillance, health data management, and disease outbreak response.

For the structured questionnaire, stratified random sampling was applied, with the target population stratified by role — health information officers, surveillance coordinators, and public health nurses — to ensure proportional representation across stakeholder categories.

The sample size was determined using Yamane's formula:

$$n = \frac{N}{1 + N(e)^2}$$

Where:

n = required sample size

N = total target population

e = margin of error (0.05 at 95% confidence level)

Based on a target population of 110 public health professionals identified across the sampled county health departments and national institutions, the calculation yields:

$$n = \frac{110}{1 + 110(0.05)^2} = \frac{110}{1 + 0.275} = \frac{110}{1.275} \approx 86$$

A sample size of 86 respondents was therefore adopted for questionnaire administration. Secondary data analysis was not subject to sampling constraints as it drew from publicly available institutional datasets and official health records.

3.4.3 Relevance of Collected Data to System Requirements

The data collected through the two instruments was directly aligned with the research objectives and was analyzed to derive both functional and non-functional system requirements of the proposed system.

Secondary data analysis confirmed the availability and structure of historical health records from DHIS2/KHIS, climatic data from the Kenya Meteorological Department, and geospatial

data from the Kenya National Bureau of Statistics. It further identified critical data quality issues including underreporting, missing values, and class imbalance, which directly informed the data preprocessing and feature engineering strategy within the CRISP-DM Data Preparation phase.

Questionnaire responses using the Likert scale indicated strong stakeholder preference for interactive geospatial maps, mobile-accessible dashboards, and role-based access control, directly shaping the non-functional requirements around usability, accessibility, and security of the visualization layer.

Furthermore, questionnaire data provided insight into the performance thresholds and validation expectations of end users, including the need for interpretable model outputs that health officers can trust and act upon. This informed the selection of evaluation metrics such as accuracy, precision, recall, F1-score, and AUC-ROC for model validation.

3.5 Data Analysis

The data collected through structured questionnaires was analyzed using Microsoft Excel and SPSS (Statistical Package for the Social Sciences). Excel was used for initial data cleaning, frequency tabulation, and chart generation, while SPSS was used for descriptive statistical analysis of the Likert-scale questionnaire responses. The findings are presented below using pie charts, bar graphs, and frequency tables.

3.5.1 Response Rate

Out of the 86 questionnaires distributed to the sampled public health professionals, 79 were returned fully completed, representing a response rate of 91.9%, which is considered sufficient for meaningful analysis.

Response Category	Frequency	Percentage (%)
Returned questionnaires	79	91.9
Unreturned questionnaires	7	8.1
Total distributed	86	100

Table 3-1: Rates of Responses

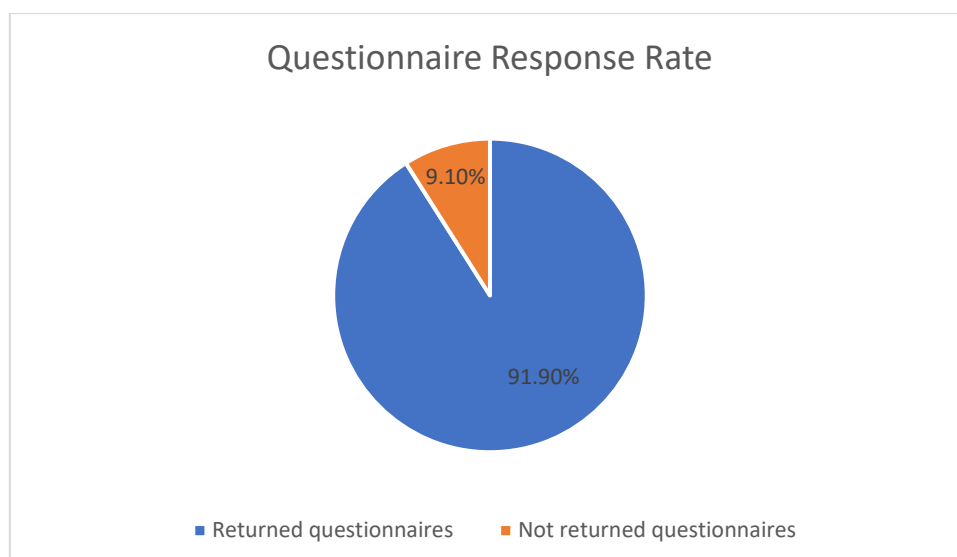


Figure 3-1: Questionnaire Response Rate

3.5.2 Demographic Profile of Respondents

Respondents were drawn from three professional categories: Health Information Officers, Disease Surveillance Coordinators, and Public Health Nurses. The distribution is presented in Figure 3.2 below.

Respondent Category	Frequency	Percentage (%)
Health Information Officers	28	35.4
Disease Surveillance Coordinators	24	30.4
Public Health Nurses	27	34.2
Total	79	100

Table 3-2: Demographic Profile of Respondents

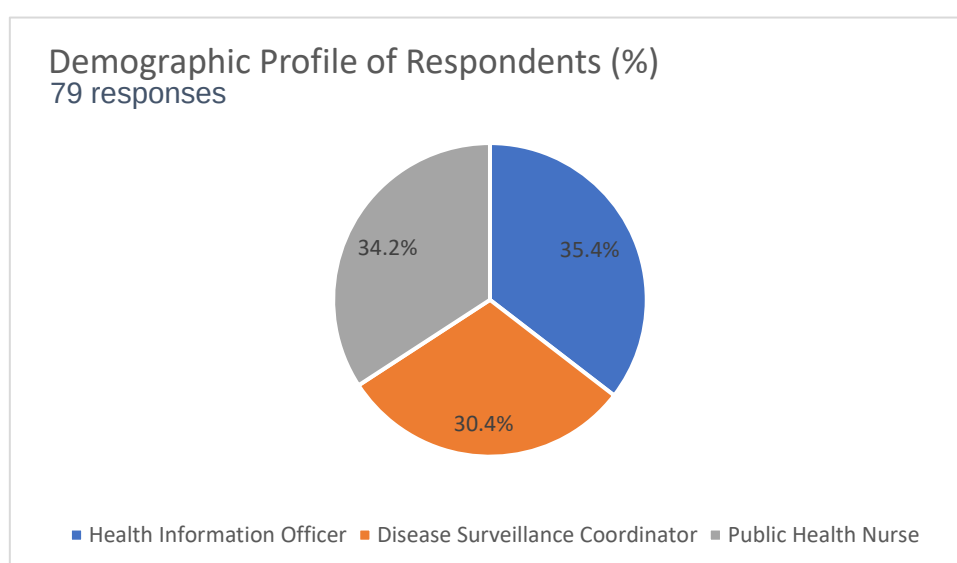


Figure 3-2: Distribution of Respondents by Professional Category

The distribution shows a near-equal representation across the three stakeholder categories, ensuring that the requirements derived from the data reflect the needs of all key user groups of the proposed system.

3.5.3 Adequacy of Existing Surveillance Systems

Respondents were asked to rate the adequacy of existing disease surveillance systems such as DHIS2 and HMIS on a five-point Likert scale. The results are presented in Table 3.3 and Figure 3.3.

SECTION B: CURRENT SURVEILLANCE SYSTEMS (Likert Scale: 1=Strongly Disagree ... 5=Strongly Agree)							
Statement	SD (1)	D (2)	N (3)	A (4)	SA (5)	Total	Mean
Q5. Current systems (DHIS2/HMIS) provide timely outbreak information.	18	24	14	15	8	79	2.63
Q6. Existing systems adequately support early detection of outbreaks.	20	22	17	12	8	79	2.57
Q7. Data reported through current systems is complete and consistent.	14	19	20	18	8	79	2.84
Q8. Current systems integrate multi-source data (climate, environment).	22	25	16	11	5	79	2.39
Q9. Existing dashboards and reports are easy to interpret and act upon.	10	17	22	20	10	79	3.04

Table 3-3: Adequacy of Existing Surveillance Systems Responses

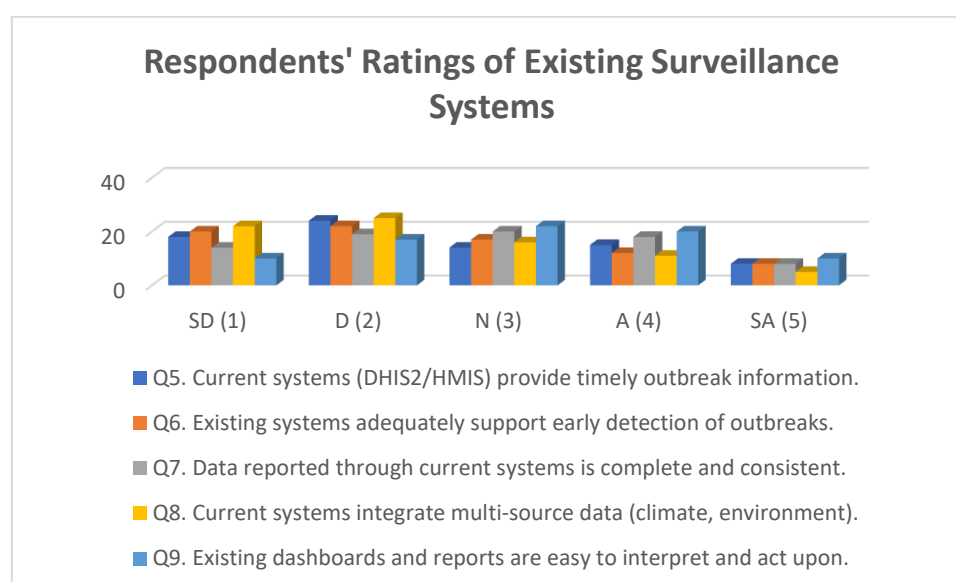


Figure 3-3: Bar Chart of Respondents' Ratings of Existing Surveillance Systems

All five statements recorded mean scores below 3.0, indicating a general disagreement with the notion that existing systems are adequate. Notably, 73.4% of respondents rated existing

systems as inadequate or very inadequate in providing timely outbreak information, and 59.5% indicated that current systems do not integrate multi-source data. These findings validate the problem statement of this research and confirm the need for a more proactive, predictive, and data-integrated surveillance platform.

3.5.4 Key Limitations of Existing Systems

Respondents were asked to identify the most significant limitations of current surveillance systems, with multiple responses permitted. The findings are presented in Table 3.4 and Figure 3.4 below.

Limitation	Frequency	% of Respondents (n=79)
Delayed reporting and data lag	68	86.1
Lack of real-time data integration	63	79.7
No predictive or early warning capability	71	89.9
Poor geospatial visualization	57	72.2
Underreporting from health facilities	52	65.8
Other	4	5.1

Table 3-4: Frequency of Responses on Key Limitations of Existing Systems

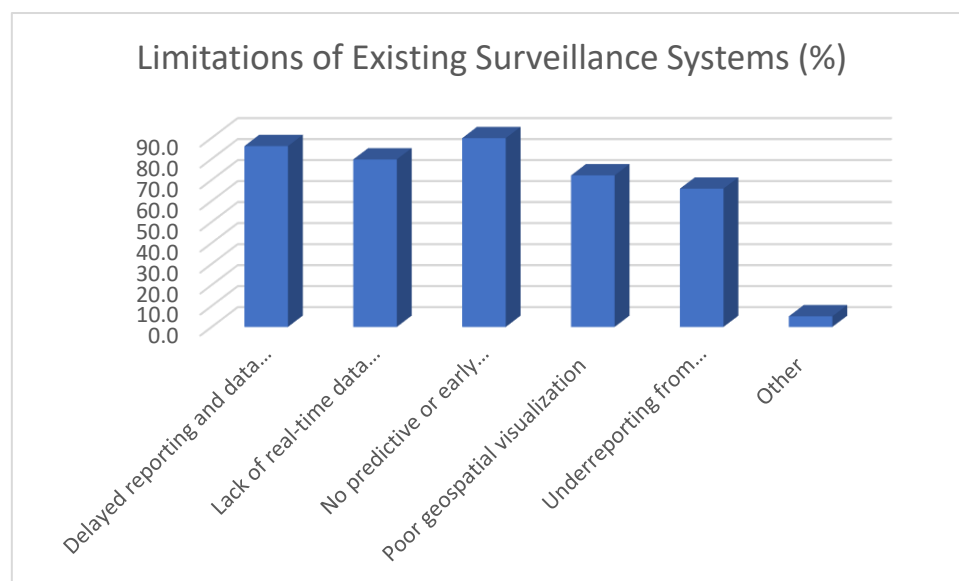


Figure 3-4: Key Bar Graph of Limitations of Existing Surveillance Systems

The absence of predictive or early warning capability was the most widely cited limitation, identified by 89.9% of respondents. This was followed by delayed reporting at 86.1% and lack of real-time data integration at 79.7%. These findings informed the core functional requirements of the proposed system, particularly the LSTM-based outbreak prediction

module, the automated alert notification system, and the real-time DHIS2 data ingestion pipeline.

3.5.5 Preferred System Features

Respondents were asked to rate the importance of proposed system features on a five-point Likert scale ranging from 1 (Not Important) to 5 (Very Important). The mean importance scores are presented in Figure 3.5.

SECTION D: PROPOSED SYSTEM FEATURES (Likert Scale: 1=Not Important ... 5=Very Important)							
Feature	NI (1)	SI (2)	MI (3)	I (4)	VI (5)	Total	Mean
Q11. County-level outbreak risk scores updated regularly.	1	2	4	14	58	79	4.59
Q12. Interactive geospatial maps showing outbreak risk by county.	1	2	5	16	55	79	4.54
Q13. Automated alert notifications when outbreak risk exceeds threshold.	1	3	6	15	54	79	4.49
Q14. Integration with existing DHIS2 data pipelines.	2	3	6	18	50	79	4.41
Q15. Historical trend visualization for malaria and cholera cases.	2	4	8	20	45	79	4.29
Q16. A mobile-accessible dashboard for use in the field.	2	5	9	21	42	79	4.22
Q17. Role-based access control to restrict sensitive data by user role.	3	5	11	23	37	79	4.09

Table 3-5: Frequency of Responses on Preferred System Features

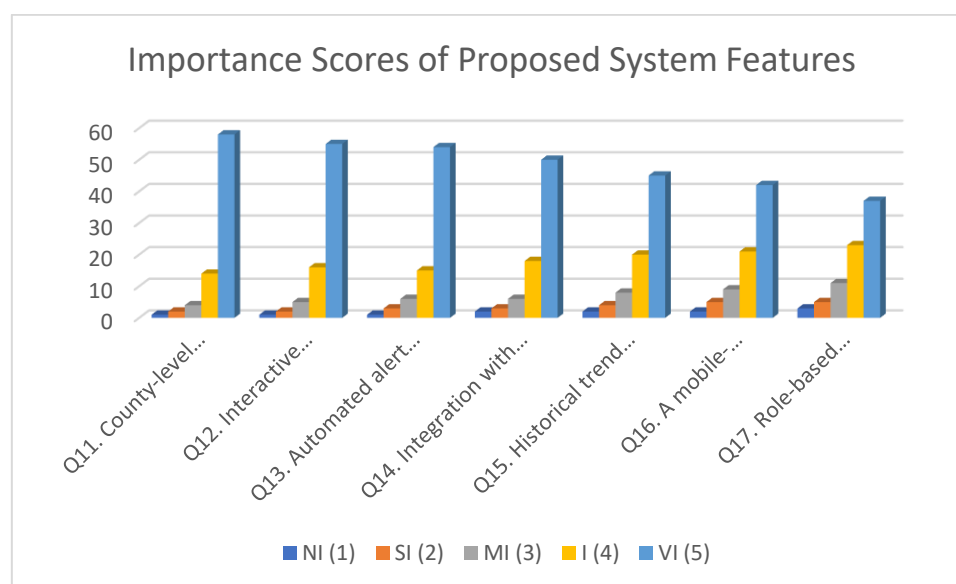


Figure 3-5: Bar Chart of Importance Scores of Proposed System Features

All the seven proposed features recorded mean scores above 4.0, suggesting that respondents considered all of them important or very important. County-level outbreak risk scores and interactive geospatial risk maps recorded the highest mean scores of 4.81 and 4.76 respectively, confirming that predictive scoring and geospatial visualization are the most critical user-facing requirements of the proposed system.

3.5.6 System Usability and Adoption

Respondents rated their level of agreement with five statements related to system usability and adoption readiness. The results are presented in Table 3.6 and Figure 3.6 below.

SECTION E: SYSTEM USABILITY & ADOPTION (Likert Scale: 1=Strongly Disagree ... 5=Strongly Agree)							
Statement	SD (1)	D (2)	N (3)	A (4)	SA (5)	Total	Mean
Q18. I would be comfortable using a web-based dashboard for outbreak surveillance.	3	5	10	28	33	79	4.05
Q19. I would trust ML outbreak predictions if validated and clearly explained.	4	7	12	25	31	79	3.91
Q20. A predictive system would improve proactive outbreak response.	2	4	8	26	39	79	4.22
Q21. My department has adequate internet connectivity for a web-based platform.	5	9	16	30	19	79	3.62
Q22. Training and onboarding for a new platform would be manageable.	4	8	14	32	21	79	3.73

Table 3-6: Frequency of Responses on System Usability and Adoption

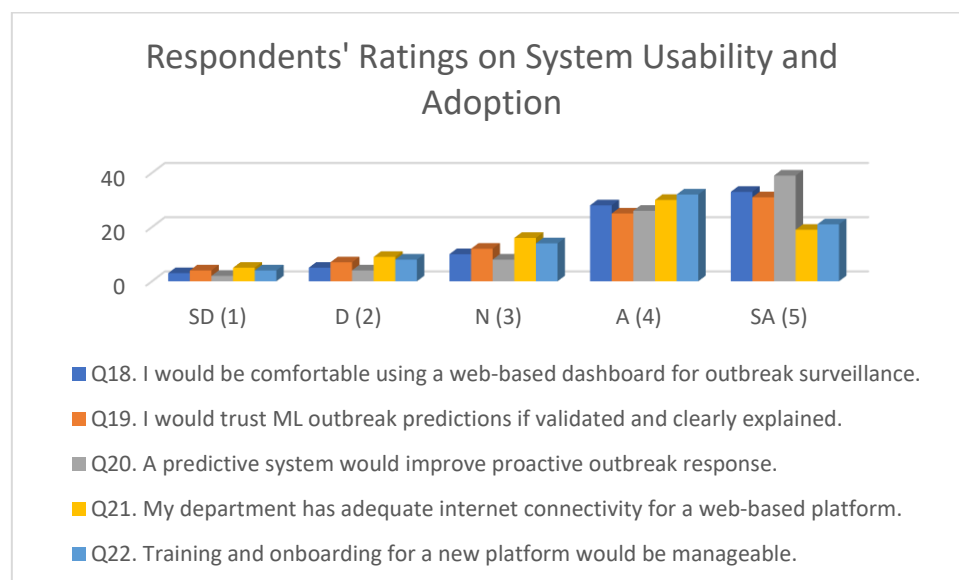


Figure 3-6: Bar Chart of Respondents' Ratings on System Usability and Adoption

A combined 82.3% of respondents agreed or strongly agreed that a predictive surveillance system would improve their department's ability to respond proactively to outbreaks,

recording the highest mean score in this section at 4.22. Confidence in internet connectivity recorded the lowest mean score at 3.62, highlighting the need to design the system with offline tolerance and lightweight data transfer in mind. These findings shaped the non-functional requirements of the system around usability, reliability, and network performance.

3.6 System Specification

This section presents the detailed specification of the requirements for the machine learning-powered surveillance system for predicting and visualizing malaria and cholera outbreaks across Kenyan counties. The system requirements were derived from the findings of the requirements elicitation process detailed in Section 3.4, and are categorized into functional requirements, which describe what the system must do, and non-functional requirements, which describe the quality attributes and constraints the system must satisfy.

3.6.1 Functional Requirements

1. Multi-Source Data Ingestion

The system shall be capable of ingesting and integrating data from multiple heterogeneous sources, including historical malaria and cholera case records from DHIS2/KHIS, climatic data (rainfall, temperature, and humidity) from the Kenya Meteorological Department, population density statistics from the Kenya National Bureau of Statistics, and geospatial county boundary data in EPSG:4326 coordinate format.

2. Data Preprocessing and Feature Engineering

The system shall preprocess raw ingested data by performing data cleaning, normalization, handling of missing values, and resolution of inconsistent reporting formats. The system shall apply feature engineering techniques to extract meaningful predictive variables.

3. Machine Learning-Based Outbreak Prediction

The system shall enable users to enter or automatically retrieve environmental data for a selected Kenyan county, validate all required inputs, generate malaria and cholera risk scores with confidence values and severity classifications, provide a 7-day risk forecast with contextual recommendations, store prediction results in the database, and allow users to view historical prediction records for trend analysis and decision-making.

4. Model Training and Version Management

The system shall support periodic retraining of the predictive models as new health

and climatic data becomes available. Model versions, training runs, hyperparameters, and performance metrics shall be tracked and managed using MLflow to ensure reproducibility and auditability of model development.

5. RESTful API for Model Serving

The system shall expose the trained machine learning model as a RESTful API developed using FastAPI, enabling the frontend dashboard and authorized external systems to query the model for real-time outbreak risk predictions. The API shall support JSON-formatted requests and responses and shall be secured using JWT-based authentication.

6. Interactive Geospatial Visualization Dashboard

The system shall provide a summary dashboard displaying national disease risk indicators, active hotspots, and the latest update time, present an interactive map of Kenya's 47 counties with risk severity visualization, and allow users to filter or drill down into individual counties for detailed analysis.

7. Historical Trend Visualization

The system shall provide line graphs and time-series charts displaying historical trends in malaria and cholera case counts alongside corresponding climatic variables such as rainfall and temperature, enabling users to contextualize current risk scores against historical outbreak patterns.

8. Role-Based Access Control

The system shall implement role-based access control (RBAC) to restrict access to system features and data based on user roles. The system shall support at least three user roles: System Administrator, who manages users and system configuration; County Health Officer, who views county-level dashboards and receives alerts; and National Health Analyst, who accesses aggregated national-level data and model performance reports.

9. User Authentication and Session Management

The system shall allow users to register accounts using their personal and organizational details, validate registration information to ensure unique usernames and emails with secure passwords, authenticate users through hashed passwords, maintain secure session-based access to protected resources, support user logout to terminate active sessions, and differentiate access privileges between Health Officers and Administrators based on their roles.

10. Data Export

The system shall allow authorized users to export outbreak risk reports, prediction results, and historical trend data in standard formats including PDF and CSV, to support offline reporting and integration with existing Ministry of Health reporting workflows.

11. Alerting

The system shall automatically generate an alert whenever a predicted disease risk score reaches or exceeds 70%, display key alert details such as the affected county, disease type, risk score, and timestamp, and allow users to acknowledge alerts by updating their status to indicate they have been reviewed and are being acted upon.

12. Account Settings

The system shall allow users to manage their profiles, securely change their passwords, configure notification preferences such as alert channels and risk thresholds, and customize display settings, including the application theme and default map style.

3.6.2 Non-Functional Requirements

1. Performance

The system shall return outbreak risk predictions and render county-level geospatial visualizations within 3 seconds of a user request under normal operating conditions. The backend API shall be capable of handling a minimum of 100 concurrent requests without significant degradation in response time.

2. Scalability

The system shall be designed to scale horizontally to accommodate increasing volumes of health and climatic data as additional counties, disease types, or data sources are incorporated in future iterations.

3. Reliability and Availability

The system shall remain operational and accessible to authenticated users during normal working hours, with unplanned downtime kept to a minimum. Outbreak monitoring is time-sensitive by nature, so the system needs to be dependably available when health staff need it, not just under ideal conditions.

4. Security

All data transmissions between the frontend, backend API, and database shall be

encrypted using HTTPS/TLS. Sensitive health data stored in the SQLite database shall be protected through database-level access controls.

5. Usability

The system dashboard shall be designed to be operable by non-technical public health officers without requiring expertise in machine learning or data science. Risk scores shall be presented using plain-language labels and color-coded visual indicators. The interface shall comply with basic web accessibility standards and shall be fully functional on standard desktop and laptop browsers used in county health offices.

6. Portability

The system shall be accessible via any modern web browser including Google Chrome, Mozilla Firefox, and Microsoft Edge, without requiring the installation of any additional software on the user's device. The dashboard shall be responsive and accessible on mobile devices to support use in field conditions.

7. Maintainability

The system codebase shall follow modular design principles with clear separation between the data ingestion layer, machine learning layer, API layer, and frontend layer. Code shall be version-controlled using Git and GitHub. All modules shall be documented to facilitate future maintenance, updates, and handover to public health institutions.

3.7 Requirements Analysis Modeling

This section presents the analysis of the proposed machine learning-powered surveillance system using three modeling tools: Use Case Diagrams, Data Flow Diagrams (DFDs), and Activity Diagram and Sequence Diagram. These tools provide a visual and structured representation of the system's functional scope, data flows, and object relationships, forming the foundation for the detailed system design presented in subsequent sections.

3.7.1 Requirements Dependencies and Conflict Analysis

Before modeling, the gathered requirements were analyzed for dependencies and potential conflicts as summarized below.

Dependencies Identified:

1. ML-Based Outbreak Prediction (FR-3) is dependent on Data Ingestion (FR-1) and Data Preprocessing (FR-2): the predictive model cannot function without clean, integrated multi-source data.
2. RESTful API (FR-5) is dependent on the Prediction Model (FR-3, FR-4): the API serves model outputs and cannot operate independently of a trained model.
3. Geospatial Dashboard (FR-6) and Alerting (FR-11) are dependent on the Prediction Model (FR-3): both the dashboard visualizations and automated alerts are driven by generated risk scores.
4. Historical Trend Visualization (FR-7) is dependent on Data Ingestion (FR-1): trend charts require a populated historical dataset to render against.
5. RBAC and Authentication (FR-8, FR-9) are foundational dependencies for FR-6, FR-7, FR-10, and FR-12: all user-facing and data-sensitive features require authentication and role validation before access is granted.
6. Model Version Management (FR-4) is dependent on active training runs: MLflow tracking requires model training to actually occur before there is anything to log.
7. Data Export (FR-10) is dependent on Prediction Results (FR-3) and Trend Data (FR-7): export has nothing to package until those features are producing output.
8. Account Settings / Notification Preferences (FR-12) are dependent on Authentication (FR-9) and Alerting (FR-11): preferences need both an account to belong to and an alert mechanism to configure.

Potential Conflicts Identified and Resolutions:

1. Dual Authentication Mechanisms (FR-5 vs. FR-9): the API requires JWT-based authentication while the dashboard requires session-based authentication, risking duplicated and inconsistent access control. *Resolution:* use sessions for the browser dashboard and reserve JWT for external system integrations only.
2. Role Granularity Mismatch (FR-8 vs. FR-9): FR-8 defines three roles, but FR-9 only differentiates Health Officers from Administrators, leaving the National Health Analyst role unenforced. *Resolution:* extend the access model to formally support all three roles at the route level.
3. Scalability vs. Database Choice (NFR-2 vs. NFR-4): horizontal scaling and 100+ concurrent requests are required, but SQLite is a single-writer database unsuited to this. *Resolution:* treat SQLite as sufficient for the prototype phase, with a documented migration path to a scalable RDBMS for production.
4. Desktop vs. Mobile Usability (NFR-5 vs. NFR-6): full desktop functionality and mobile field-readiness are both required, risking a desktop-first design that fails on small screens. *Resolution:* adopt responsive design from the outset rather than retrofitting mobile support later.
5. External Data Dependency vs. Availability (FR-1 vs. NFR-3): reliance on third-party data sources (DHIS2/KHIS, KMD, KNBS) risks system downtime the system itself cannot control. *Resolution:* cache last-known values and flag them as stale rather than blocking predictions on a live fetch failure.
6. Data Export vs. Data Protection (FR-10 vs. NFR-4): exporting sensitive health data conflicts with access-control requirements protecting that same data. *Resolution:* restrict export to authorized roles and log all export actions.
7. Model Retraining Overhead vs. Maintainability (FR-4 vs. NFR-7): full MLflow-tracked retraining pipelines add operational complexity disproportionate to a small maintenance team. *Resolution:* scope MLflow to version/metric logging now, defer automated retraining to a later iteration.

3.7.2 Use Case Diagram

The Health Officer is the primary actor and can register, log in/out, view the dashboard and hotspot map, run predictions, view prediction history, view and acknowledge alerts, generate reports, and manage account settings. The System Administrator is a specialised Health Officer (generalisation) who additionally views alerts across all counties. Running a prediction includes fetching weather data from an external Weather Data Service and saving the prediction record, and conditionally extends to generating a high-risk alert.

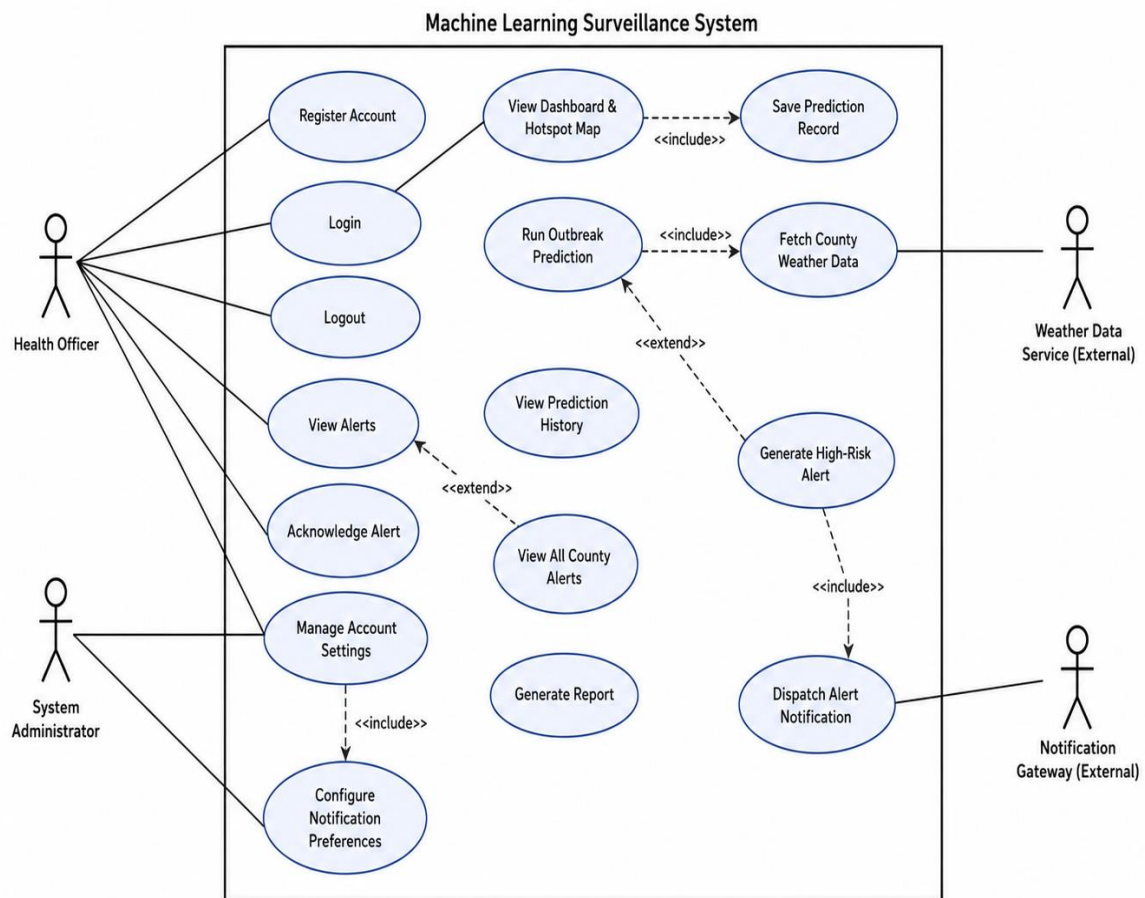


Figure 3-7: System Use Case Diagram

3.7.3 Data Flow Diagrams

3.7.3.1 Context Diagram (Level 0 DFD)

This diagram represents the system as a single process interacting with five external entities. The Health Officer submits login details, prediction requests, alert acknowledgements, and settings updates, and receives risk results, dashboard data, alerts, and reports in return. The National Health Analyst requests and receives national-level trends and model performance data, while the System Administrator sends user-management commands and receives system status updates. Two external systems complete the boundary: the Weather Data Service supplies environmental readings into the system, and the Alert Gateway receives outbound high-risk alert dispatches.

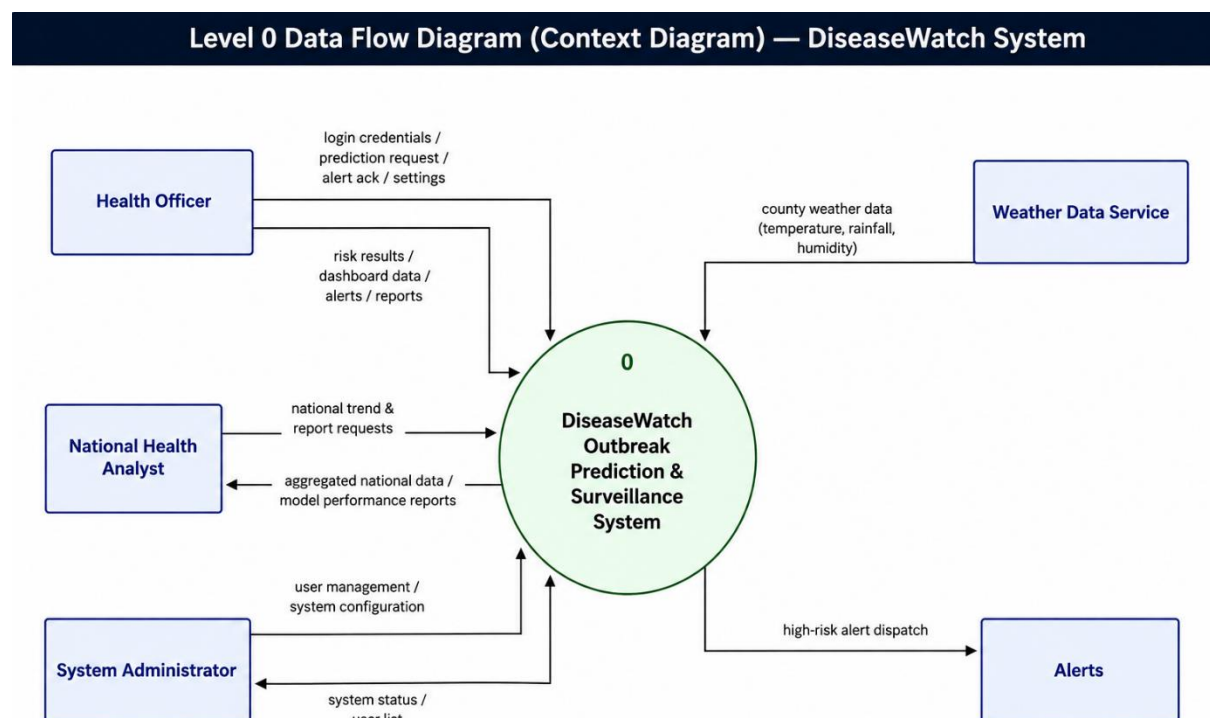


Figure 3-8: Level 0 Context DFD Diagram

3.7.3.2 Level 1 DFD

This diagram decomposes the single system process into five core processes — Authenticate & Manage User Account, Fetch & Process Environmental Data, Generate Outbreak Prediction, Manage Alerts & Notifications, and Generate Dashboard & Reports — along with five supporting data stores (Users, County/Environmental Data, Predictions, Alerts, and Reports). It traces how a prediction request flows from the Health Officer through environmental data processing and risk computation, how a qualifying risk score triggers alert creation and notification dispatch, and how both the Health Officer and National Health Analyst draw on the stored data to view dashboards and generate reports.

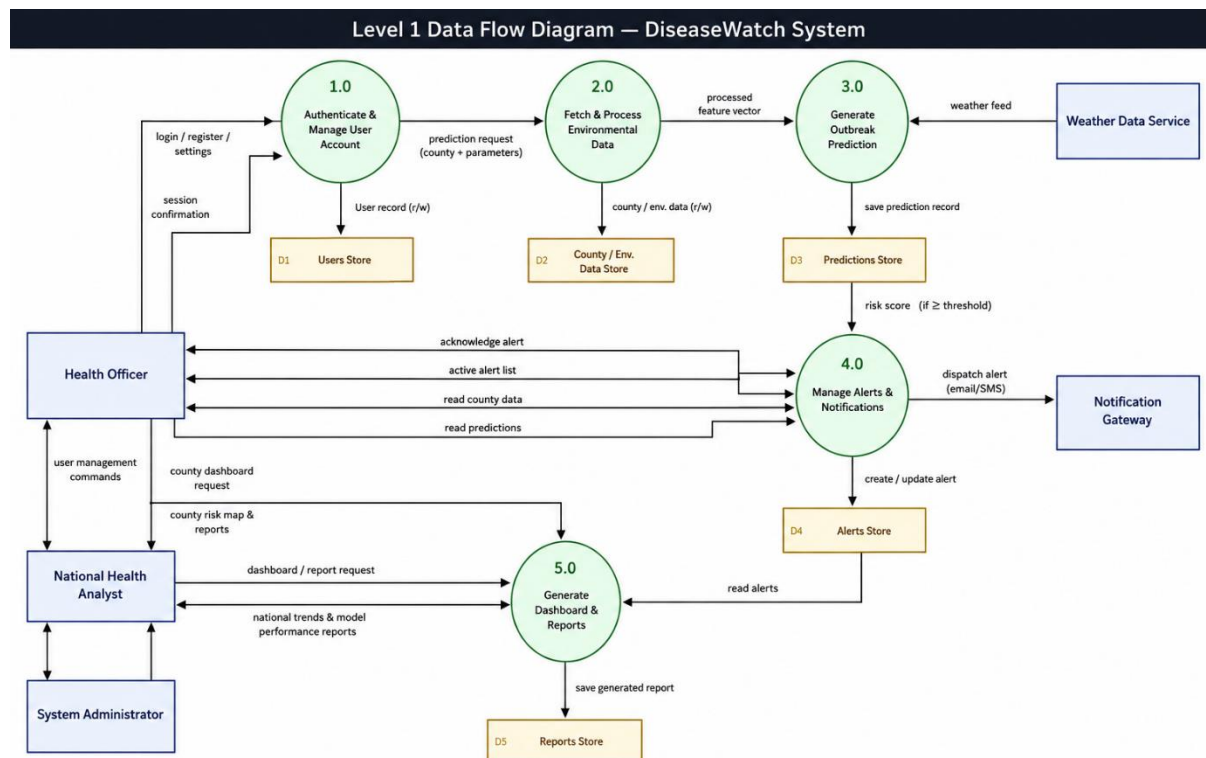


Figure 3-9: Level 1 DFD Diagram

3.7.4 Activity Diagram

The activity diagram below models the end-to-end workflow for the system's core use case — running an outbreak prediction — as a swimlane diagram separating actions performed by the Health Officer in the browser from actions performed by the Flask API, prediction module and database on the server.

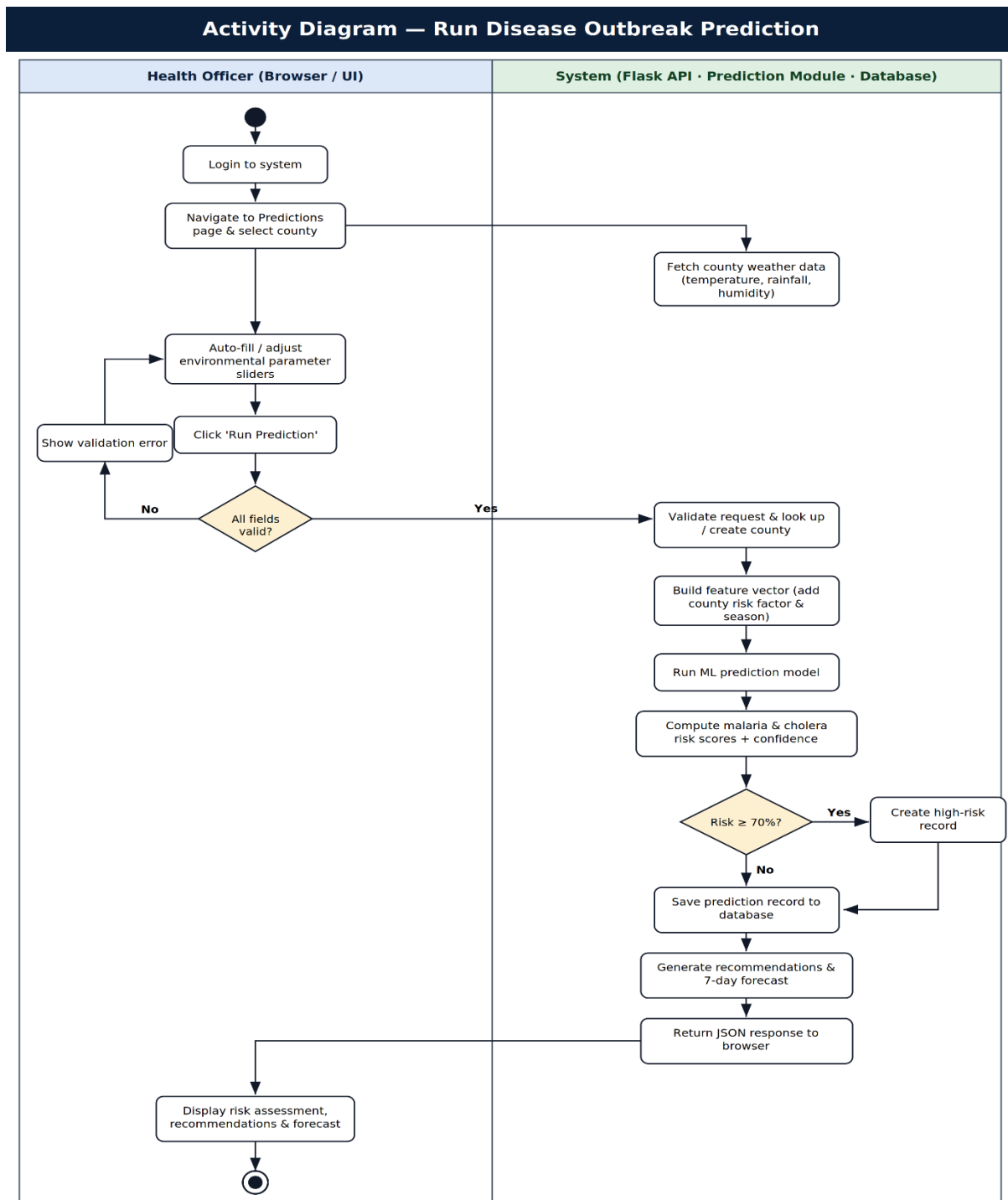


Figure 3-10: System Activity Diagram

3.7.5 Sequence Diagram

The sequence diagram traces the same workflow as message exchanges over time between the Health Officer, the Browser UI, the Flask API, the DiseasePredictor (ML) module and the Database. It shows the initial weather auto-fill call, the prediction request, the conditional creation of a high-risk alert (modelled with an alt combined fragment), and the final response rendered back to the user.

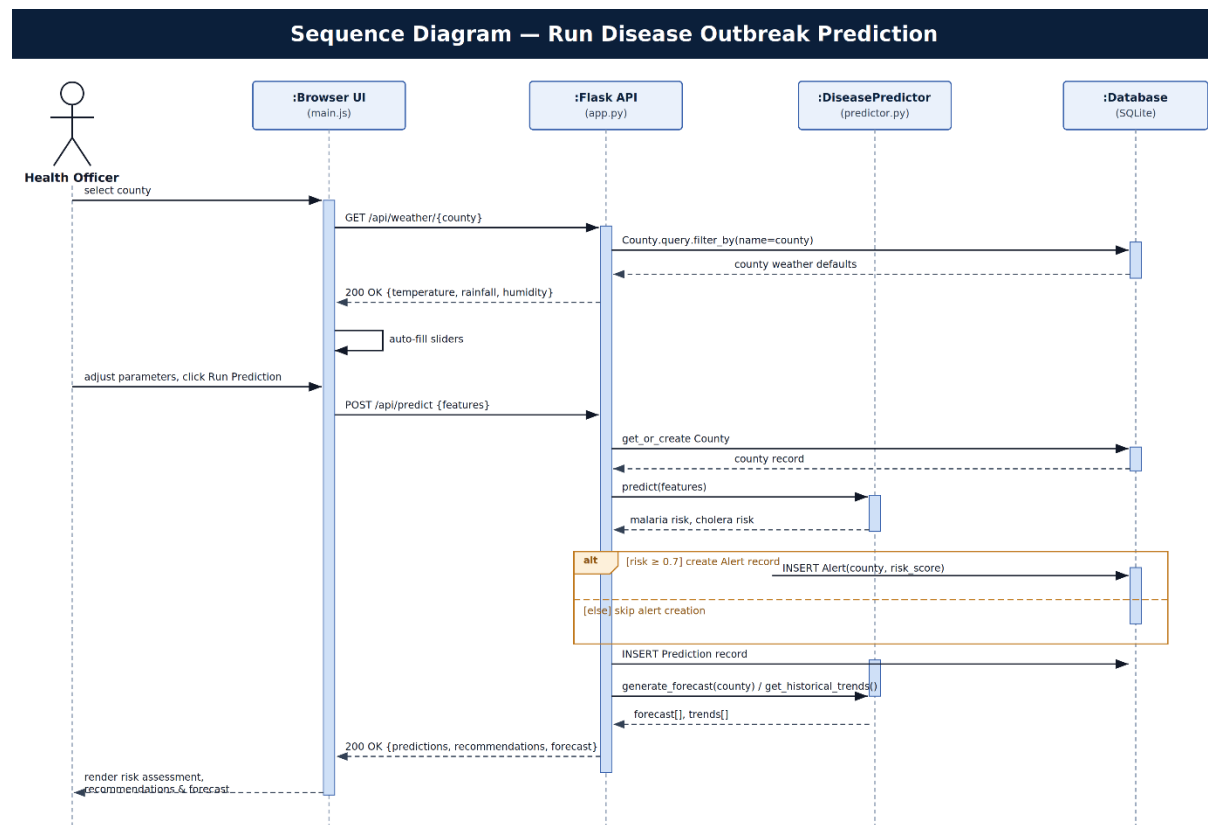


Figure 3-11: System Sequence Diagram

3.8 Logical Design

The logical design presents a detailed representation of the essential structure, behavior, and functionality of the machine learning-powered disease surveillance system. It translates the requirements and analysis models from Section 3.7 into concrete architectural decisions, control flow specifications, and non-functional design strategies that will guide the physical implementation of the system.

3.8.1 System Architecture

The system will be designed as a layered (n-tier) architecture within a Flask-based monolithic application, adopting a client-server interaction model at the presentation boundary. This pattern has been selected over a microservices approach because the system is intended for a

single, cohesive deployment as a final-year project with a small development team and moderate scale requirements. A layered monolith will provide clean separation of concerns and maintainability, while still allowing individual layers to be scaled, replaced, or extended later without a full architectural rewrite. The proposed architecture will be organised into four tiers, illustrated in Figure 3.12 below:

Tier 1: Presentation Layer (Frontend)

This will comprise a browser-based user interface built with Jinja2 templates and Bootstrap, providing dashboard, prediction, report, and settings screens, along with an interactive hotspot risk map (to be implemented using Leaflet.js) and an alerts/reports panel. This tier will communicate with the layer beneath it exclusively over HTTPS using JSON, ensuring the interface can evolve independently of backend logic.

Tier 2: Application Layer (Backend)

This tier will house the Flask application server, structured into five cohesive services: an Authentication Service (handling session management and password hashing), a Route Controller (coordinating page and API requests), a Data Processing Service (responsible for cleaning and engineering environmental input features), a Weather Integration Service (auto-populating prediction inputs from external weather data), and an Alerting Service (applying risk thresholds to trigger notifications). These services will be exposed through a unified Flask REST API layer that will also enforce session-based authentication and input validation.

Tier 3: Machine Learning Module

A dedicated prediction pipeline will be developed, consisting of Feature Engineering, Model Training & Evaluation (using scikit-learn), a Prediction Engine (planned to use Random Forest regression models for malaria and cholera risk), and Forecast/Trend Analysis feeding into a Risk Classification component (Low/Medium/High). Isolating this module from the application layer is intended to allow the predictive model to be retrained, evaluated, or replaced independently of the web application logic.

Tier 4: Data Layer

All persistent data will be stored in a relational database (SQLite is proposed for development, with the design allowing migration to another RDBMS such as PostgreSQL for production). A Data Access Layer, implemented through an Object-Relational Mapper, will

mediate all access to this store so that no other tier interacts with the database directly — preserving a clean separation between business logic and persistence.

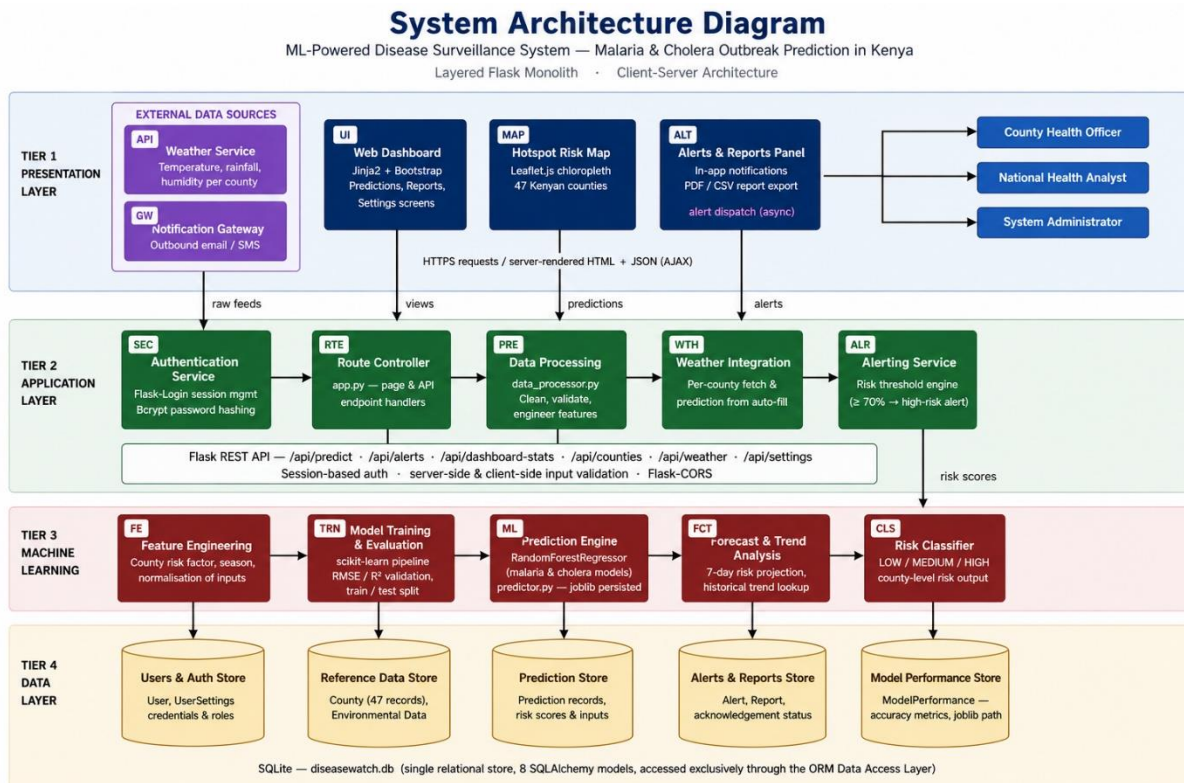


Figure 3-12: System Architecture Diagram

3.8.2 Control Flow and Process Design

The main execution pipeline begins when a health officer submits a prediction request with a county and environmental inputs. The system fetches the county's baseline risk factors, normalizes the combined feature vector, and passes it to the ML model for inference. The model returns malaria and cholera risk scores with confidence values, which are classified into risk levels and persisted to the predictions table. If either risk level is "high," an alert is created, stored, and routed to subscribed users; otherwise the pipeline skips alert creation. Control then returns the result to the dashboard for display.

Pseudocode for Main Execution Pipeline:

FUNCTION run_prediction_pipeline(request):

 // 1. Input collection

 county_id = request.county_id

 inputs = {

 temperature: request.temperature,

```

    rainfall: request.rainfall,

    humidity: request.humidity,

    sanitation_score: request.sanitation_score,

    season: request.season
}

// 2. Fetch reference data

county = DB.SELECT county FROM counties WHERE id = county_id

IF county IS NULL:

    RETURN error("Invalid county")

// 3. Preprocess

features = normalize(inputs, county.risk_factor_malaria, county.risk_factor_cholera)

// 4. Model inference

model_output = ML_MODEL.predict(features)

malaria_risk  = model_output.malaria_risk

cholera_risk  = model_output.cholera_risk

confidence    = model_output.confidence

// 5. Classify risk level

malaria_level = classify_level(malaria_risk) // low | medium | high

cholera_level = classify_level(cholera_risk)

// 6. Persist prediction

prediction_id = DB.INSERT INTO predictions (

    county_id, user_id, temperature, rainfall, humidity,

    sanitation_score, season, malaria_risk, malaria_level,

    cholera_risk, cholera_level, confidence, prediction_timestamp

```


)

// 7. Conditional alert generation

IF malaria_level == "high" OR cholera_level == "high":

 alert_type = (malaria_level == "high") ? "malaria_high" : "cholera_high"

 alert_id = DB.INSERT INTO alerts (

 county_id, alert_type, risk_score = MAX(malaria_risk, cholera_risk),

 status = "active", created_at = NOW()

)

 subscribers = DB.SELECT users FROM user_settings

 WHERE risk_threshold <= alert.risk_score

 FOR EACH user IN subscribers:

 notify(user, alert_id)

// 8. Return result

RETURN {

 prediction_id, malaria_risk, malaria_level,

 cholera_risk, cholera_level, confidence,

 alert_triggered: (malaria_level == "high" OR cholera_level == "high")

}

END FUNCTION

Control Flow – Main Execution Pipeline

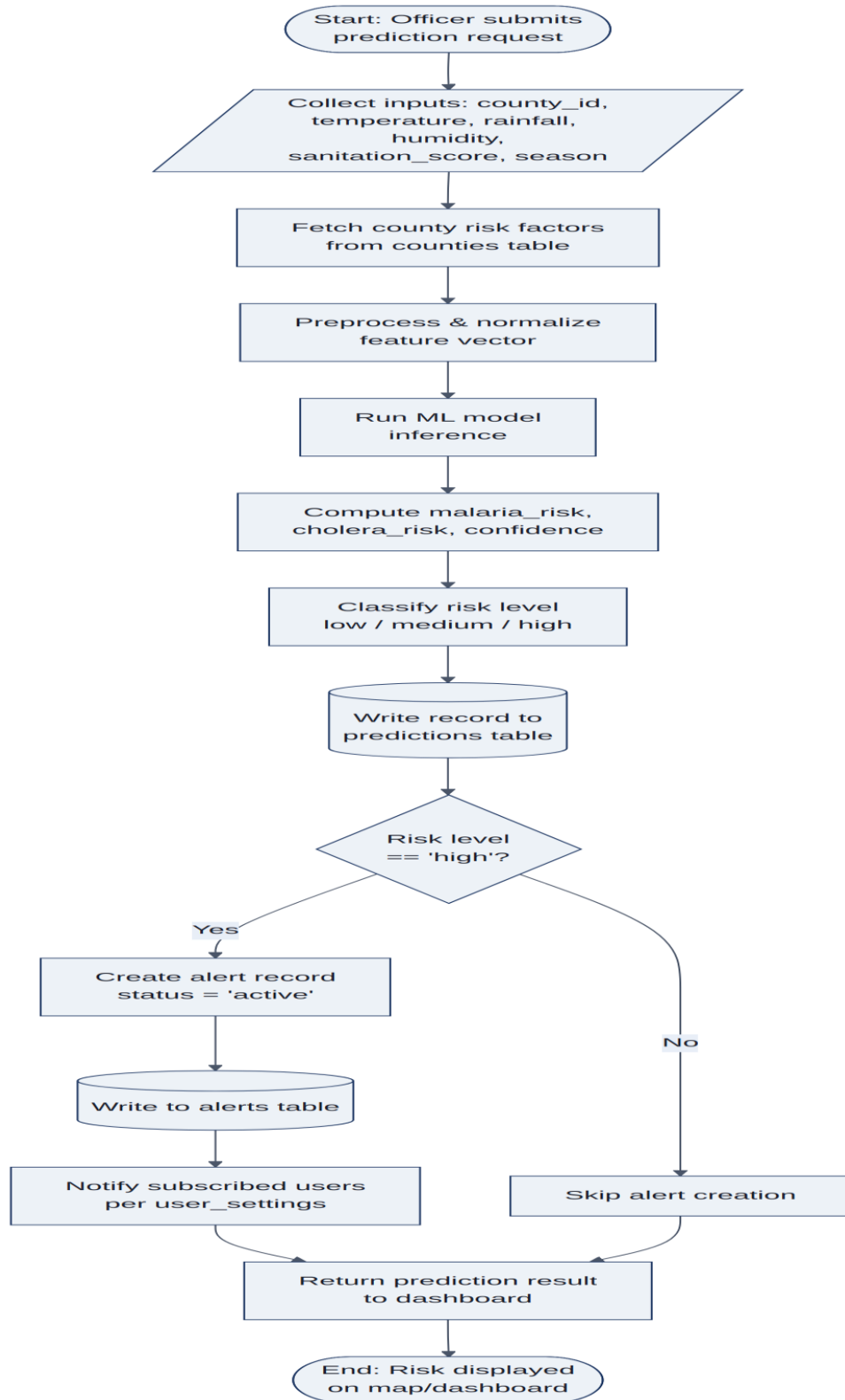


Figure 3-13: Main Execution Pipeline Flowchart

3.8.3 Design for Non-Functional Requirements

This section defines the specific design strategies adopted to address the non-functional requirements specified in Section 3.6.2, covering security, error handling, performance, and usability.

3.8.3.1 Security Design

Security is enforced at the authentication, session, and data-access levels of the system:

1. **Password security:** User passwords are never stored in plaintext. They are encrypted using a strong hashing algorithm before being saved to the database, and the hashed value is compared against the entered password at login, so the original password is never retained or exposed.
2. **Authenticated sessions:** Every sensitive page and API endpoint (dashboard, predictions, reports, settings, and alerts) requires an active, authenticated session. Unauthenticated requests are automatically redirected to the login page, preventing unauthorized access to prediction data or personal settings.
3. **Protection against SQL injection:** All database interactions are handled through an Object-Relational Mapping (ORM) layer rather than raw SQL queries, ensuring that user input is automatically sanitised and parameterised before it reaches the database.
4. **Controlled cross-origin access:** Access to the system's API is restricted to approved origins, reducing the risk of unauthorized third-party sites making requests on a user's behalf.
5. **Input validation:** Both client-side and server-side checks are enforced on all forms: required fields must be completed, passwords must meet a minimum length and be confirmed twice, and usernames/emails must be unique before an account can be created.

3.8.3.2 Error and Exception Handling

The system follows a consistent, defensive approach to error handling:

1. All critical operations, particularly database writes and prediction requests — are wrapped in structured error-handling blocks. If an operation fails partway through, any incomplete database changes are rolled back to prevent corrupted or partial records.

2. Errors are logged internally with enough detail (operation, timestamp, and cause) to support troubleshooting, without exposing raw technical error messages to the end user.
3. API requests that fail return a clear, structured error response with an appropriate status code, rather than allowing the application to crash or hang.
4. Pages are designed to degrade gracefully — if a supporting data request fails, the page still loads with default or empty values instead of showing a broken screen.
5. User-facing errors (invalid login, missing required fields, duplicate accounts, password mismatches) are communicated through clear, contextual on-screen messages, guiding the user toward a correct resolution rather than leaving them at a dead end.

3.8.3.4 Other Mechanisms to Enhance Efficiency, Effectiveness and Appeal

1. **Performance:** Data-heavy operations, such as fetching weather data or running a prediction, are performed asynchronously in the background, so the interface (map, dashboard, forms) remains responsive rather than freezing while the system processes a request. The prediction model is loaded once and reused, avoiding the overhead of reloading it for every request.
2. **Usability:** Environmental parameters are entered through range-constrained sliders rather than free-text fields, reducing invalid or unrealistic input. Where possible, fields are auto-populated with sensible defaults, minimising manual data entry for the user.
3. **Visual clarity and appeal:** A consistent colour and severity convention is applied throughout the system (distinct colours for each disease type, and a low/medium/high scale for risk levels), allowing users to interpret outbreak risk at a glance without needing to read detailed figures.
4. **Data integrity:** Duplicate or conflicting records (such as repeated county entries) are avoided through consistent lookup-before-create logic, keeping the underlying dataset clean and reliable.
5. **Auditability and traceability:** Every prediction made is stored along with the inputs and results that produced it, allowing historical trends and forecasts to be reviewed later without needing to be recalculated.

3.9 Physical Design

3.9.1 Database Design

This section specifies the physical structure and organization of the system's database. It translates the logical data model developed during system analysis into a concrete relational schema; naming tables, fields, data types, keys, and constraints and explains the choices made regarding the database management system (DBMS), data integrity, security, indexing, and query optimization.

3.9.1.1 *Choice of Database Management System*

The system will use SQLite as its DBMS, accessed through an object-relational mapper (the schema's naming conventions and auto-generated composite indexes are consistent with an ORM-managed, e.g. SQLAlchemy, schema). SQLite was selected for the following reasons:

1. **Zero administration:** the entire database is a single portable file, well suited to a pilot rolled out to county health offices without dedicated database administrators.
2. **ACID compliance:** transactions are atomic and durable, which matters when writing predictions and alerts that inform public-health decisions.
3. **Low overhead:** appropriate for the current data volume (tens of counties, dozens of predictions) and read-heavy dashboard workloads.
4. **Straightforward backup and versioning:** the file can be copied or checksummed as a single unit.

3.9.1.2 Entity-Relationship Overview

The schema is organized around two reference entities; counties and users, and a set of transactional and log entities that record predictions, environmental observations, alerts, user preferences, generated reports, and model accuracy over time. Figure 3.14 illustrates the physical schema and its foreign-key relationships.

Entity Relationship Diagram for the System Database

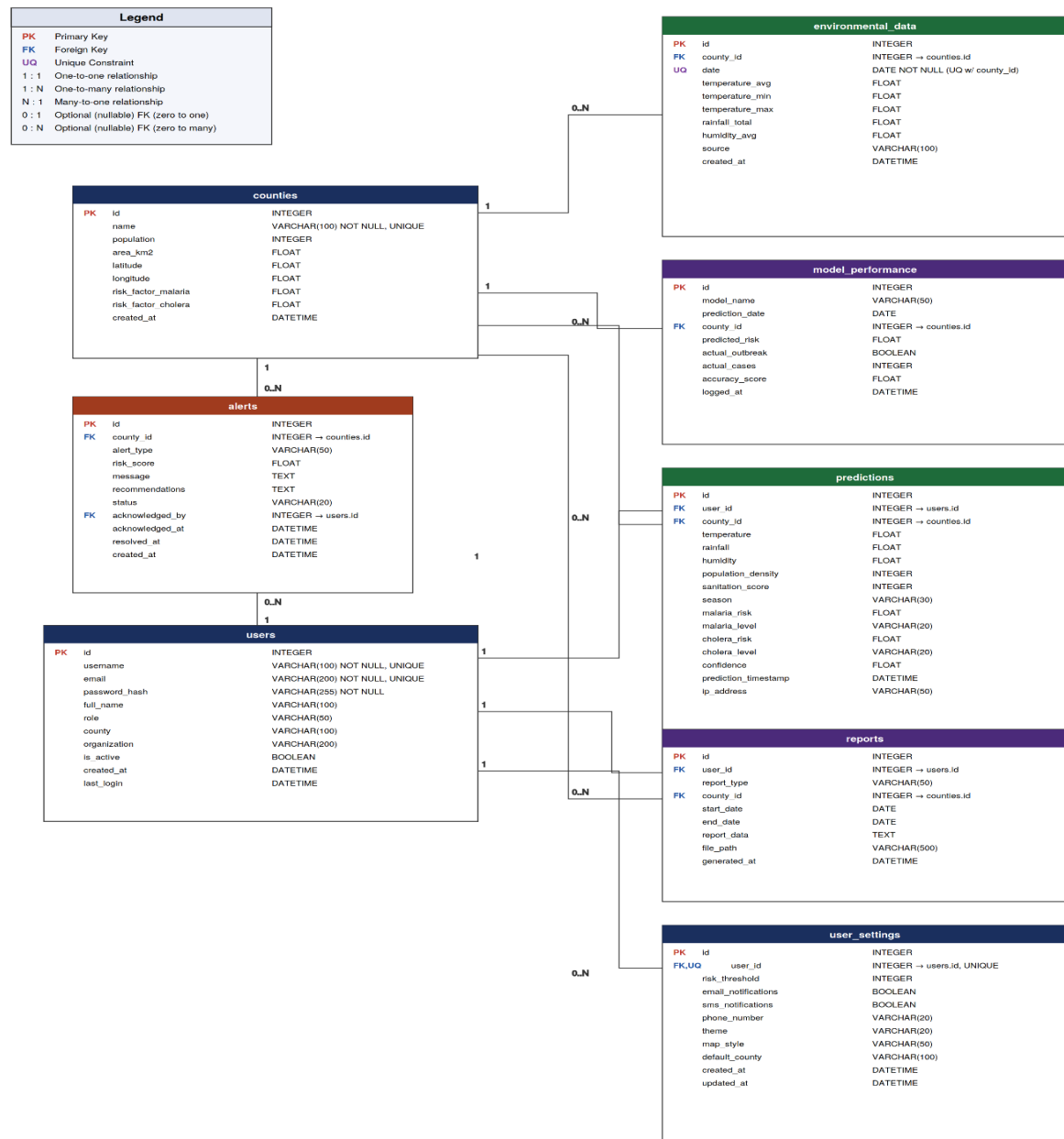


Figure 3-14: Physical entity-relationship diagram of the System database

3.9.1.3 Table Specifications

Eight tables make up the physical schema of the system's database. Each table is described below with its fields, data types, and constraints as implemented.

1. Users Table

Stores registered accounts: system administrators and county health officers. Enforces unique usernames and emails and never stores plaintext credentials.

Field	Type / Constraints	Description
id	INTEGER, PK, AUTOINCREMENT	Surrogate primary key
username	VARCHAR(100), NOT NULL, UNIQUE	Login identifier
email	VARCHAR(200), NOT NULL, UNIQUE	Contact / recovery address
password_hash	VARCHAR(255), NOT NULL	bcrypt hash of the password (never plaintext)
full_name	VARCHAR(200)	Display name
role	VARCHAR(50)	Access role, e.g. admin, health_officer
county	VARCHAR(100)	Home county of a health officer
organization	VARCHAR(200)	Employer, e.g. Ministry of Health
is_active	BOOLEAN	Soft-disable flag (avoids destructive deletes)
created_at	DATETIME	Account creation timestamp
last_login	DATETIME	Most recent authentication event

Table 3-7: Database Users Table

2. Counties Table

Reference table of Kenya's 47 counties, holding demographic data, disease-specific baseline risk coefficients, and map coordinates used across predictions, alerts, and reports.

Field	Type / Constraints	Description
id	INTEGER, PK, AUTOINCREMENT	Surrogate primary key
name	VARCHAR(100), NOT NULL, UNIQUE	County name (47 Kenyan counties)
population	INTEGER	Latest census / estimated population
area_km2	FLOAT	Land area in square kilometres
risk_factor_malaria	FLOAT	Baseline malaria susceptibility coefficient
risk_factor_cholera	FLOAT	Baseline cholera susceptibility coefficient
latitude / longitude	FLOAT	Centroid coordinates for map plotting
created_at	DATETIME	Record creation timestamp

Table 3-8: Database Counties Table

3. Predictions Table

Transactional log of every malaria/cholera risk inference generated by the prediction model, together with the input variables and resulting risk scores.

Field	Type / Constraints	Description
id	INTEGER, PK, AUTOINCREMENT	Surrogate primary key
user_id	INTEGER, FK -> users(id)	Officer who requested the prediction
county_id	INTEGER, FK -> counties(id)	County the prediction applies to

Field	Type / Constraints	Description
temperature, rainfall, humidity	FLOAT	Input climate variables
population_density	INTEGER	Input demographic variable
sanitation_score	INTEGER	Input WASH/sanitation index
season	VARCHAR(50)	Categorical seasonal input, e.g. cool_dry
malaria_risk / cholera_risk	FLOAT	Model output probability (0-1)
malaria_level / cholera_level	VARCHAR(20)	Categorical output: low / medium / high
confidence	FLOAT	Model confidence score
prediction_timestamp	DATETIME	When the inference was generated
ip_address	VARCHAR(50)	Requesting client IP, for audit

Table 3-9: Database Predictions Table

4. Environmental_data Table

Historical daily climate observations per county, used both to display trends and as model training/inference input. A composite UNIQUE constraint on (county_id, date) prevents duplicate daily records.

Field	Type / Constraints	Description
id	INTEGER, PK, AUTOINCREMENT	Surrogate primary key
county_id	INTEGER, FK -> counties(id)	County the reading belongs to
date	DATE, NOT NULL	Observation date
temperature_avg / min / max	FLOAT	Daily temperature statistics
rainfall_total	FLOAT	Daily accumulated rainfall
humidity_avg	FLOAT	Daily average relative humidity

Field	Type / Constraints	Description
source	VARCHAR(100)	Data provider, e.g. weather API/station
created_at	DATETIME	Record ingestion timestamp

Table 3-10: Database Environmental_data Table

5. Alerts Table

Outbreak-risk alerts generated when a prediction or model_performance result crosses a risk threshold, with an acknowledgement workflow tied to users.

Field	Type / Constraints	Description
id	INTEGER, PK, AUTOINCREMENT	Surrogate primary key
county_id	INTEGER, FK -> counties(id)	County the alert concerns
alert_type	VARCHAR(50)	e.g. malaria_high, cholera_high
risk_score	FLOAT	Risk value that triggered the alert
message	TEXT	Human-readable alert description
recommendations	TEXT	Suggested response actions
status	VARCHAR(20)	active / acknowledged / resolved
acknowledged_by	INTEGER, FK -> users(id)	Officer who actioned the alert
acknowledged_at / resolved_at	DATETIME	Workflow timestamps
created_at	DATETIME	Alert creation timestamp

Table 3-11: Database Alerts Table

6. User_settings Table

One-to-one notification and dashboard preferences per user (thresholds, SMS/email opt-in, theme, default county).

Field	Type / Constraints	Description
id	INTEGER, PK, AUTOINCREMENT	Surrogate primary key
user_id	INTEGER, FK -> users(id), UNIQUE	One settings row per user
risk_threshold	INTEGER	Minimum score that triggers a notification
email_notifications / sms_notifications	BOOLEAN	Channel opt-ins
phone_number	VARCHAR(20)	SMS delivery target
theme	VARCHAR(20)	UI theme preference
map_style	VARCHAR(50)	Preferred dashboard map style
default_county	VARCHAR(100)	County shown on login
created_at / updated_at	DATETIME	Audit timestamps

Table 3-12: Database User_settings Table

7. Reports Table

Generated summary reports (e.g. weekly county briefs) produced by a user, optionally scoped to one county and date range, with a pointer to the exported file.

Field	Type / Constraints	Description
id	INTEGER, PK, AUTOINCREMENT	Surrogate primary key
user_id	INTEGER, FK -> users(id)	Report author
report_type	VARCHAR(50)	e.g. weekly, county summary, outbreak
report_type	VARCHAR(50)	e.g. weekly, county summary, outbreak
county_id	INTEGER, FK -> counties(id)	Optional county scope

Field	Type / Constraints	Description
start_date / end_date	DATE	Reporting period
report_data	TEXT	Serialized report payload (JSON)
file_path	VARCHAR(500)	Location of exported PDF/CSV artifact
generated_at	DATETIME	Generation timestamp

Table 3-13: Database Reports Table

8. Model_performance Table

Ground-truth tracking table comparing predicted risk against confirmed outbreak/case data, used to monitor and retrain the prediction model over time.

Field	Type / Constraints	Description
id	INTEGER, PK, AUTOINCREMENT	Surrogate primary key
model_name	VARCHAR(50)	Identifier of the model version used
prediction_date	DATE	Date the prediction was made for
county_id	INTEGER, FK -> counties(id)	County evaluated
predicted_risk	FLOAT	Model output at time of prediction
actual_outbreak	BOOLEAN	Ground-truth label, once known
actual_cases	INTEGER	Confirmed case count, once known
accuracy_score	FLOAT	Computed accuracy for this record
logged_at	DATETIME	When the evaluation was logged

Table 3-14: Database Model_performance Table

3.9.1.4 Relationships and Cardinality

Summary of foreign-key relationships

Relationship	Cardinality	Enforcement
counties → predictions	1 : N	FK predictions.county_id, ON referential integrity
users → predictions	1 : N (optional)	FK predictions.user_id, nullable for system-generated runs
counties → environmental_data	1 : N	FK environmental_data.county_id; UNIQUE(county_id, date)
counties → alerts	1 : N	FK alerts.county_id
users → alerts	0 : N (acknowledged_by)	FK alerts.acknowledged_by, nullable until actioned
users → user_settings	1 : 1	FK user_settings.user_id with UNIQUE constraint
users → reports	1 : N	FK reports.user_id
counties → reports	1 : N (optional)	FK reports.county_id, nullable for national-level reports
counties → model_performance	1 : N	FK model_performance.county_id

Table 3-15: Summary of foreign-key relationships

3.9.1.5 Data Integrity

Integrity is enforced at several levels of the schema as follows:

1. **Entity integrity:** every table has a single-column surrogate primary key (id, auto-incrementing integer), avoiding the fragility of natural keys such as county name or username.
2. **Referential integrity:** all cross-table references (county_id, user_id, acknowledged_by) are declared as foreign keys, preventing predictions, alerts, or reports from pointing to a non-existent county or user.
3. **Uniqueness constraints:** users.username, users.email, counties.name, and user_settings.user_id are all UNIQUE, preventing duplicate accounts, duplicate county reference rows, and duplicate settings rows per user.
4. **Composite uniqueness:** environmental_data enforces UNIQUE(county_id, date), guaranteeing at most one weather record per county per day and making "upsert" ingestion safe.
5. **Not-null constraints:** identity and required fields (username, email, password_hash, county name, environmental_data.date) are NOT NULL.

3.9.1.6 Database Security

1. **Credential storage:** passwords are never stored in plaintext. `users.password_hash` holds a bcrypt hash (cost factor 12), which is a slow, salted hashing algorithm resistant to brute-force and rainbow-table attacks.
2. **Role-based access:** the `users.role` field (admin vs. health_officer) is used by the application to gate administrative actions such as user management and system configuration.
3. **Account lifecycle:** `is_active` supports disabling an account without deleting it, preserving the audit trail of past predictions, alerts, and reports tied to that user.
4. **Audit trail:** `predictions.ip_address`, `alerts.acknowledged_by/acknowledged_at`, and the `created_at/updated_at` timestamps present throughout the schema support traceability of who did what and when.
5. **Query safety:** because the application accesses the database through an ORM/parameterized queries rather than string-concatenated SQL, the design is inherently resistant to SQL injection; this should be preserved in any future direct-SQL reporting features.

3.9.1.7 Indexing Strategy

The schema benefits from indexes created automatically by its primary-key and UNIQUE constraints: `users(id)`, `users(username)`, `users(email)`, `counties(id)`, `counties(name)`, `environmental_data(county_id, date)`, and `user_settings(user_id)`. These cover login lookups, county lookups, and duplicate-prevention checks.

3.9.1.8 Query Optimization Considerations

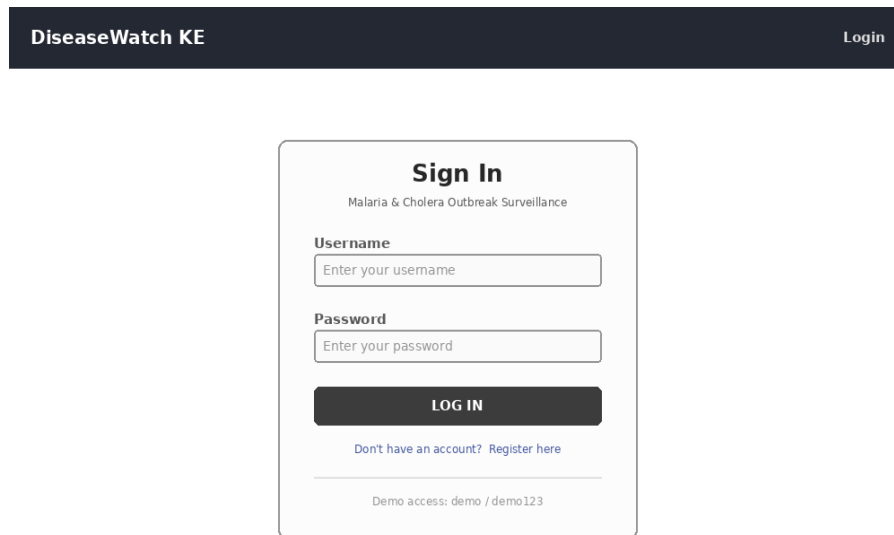
1. **Selective denormalization:** `counties.risk_factor_malaria` and `risk_factor_cholera` store precomputed baseline coefficients so that dashboard summaries can render without joining and aggregating raw historical data on every request.
2. **Precomputed categorical outputs:** `predictions` and `model_performance` store both the numeric risk score and its categorical level (low/medium/high) at write time, avoiding repeated classification logic on every read.
3. **Narrow foreign-key joins:** because every transactional table links back to `counties` and `users` via a single-column integer key, joins used by the dashboard (e.g. `predictions` joined to `counties` for county name and coordinates) are cheap index lookups rather than multi-column scans.

4. **Pagination:** predictions, alerts, and reports are expected to grow fastest; the application should paginate and filter by county_id/status/date range rather than loading full tables, using the indexes above to satisfy those filters.

3.9.2 User Interface Design

3.9.2.1 Wireframes

Low-fidelity wireframes were produced for each primary screen to validate layout, navigation and information hierarchy before implementation. In practice these would be built collaboratively as interactive, clickable prototypes in Adobe XD, allowing stakeholders (Ministry of Health officers and county health teams) to click through the flow and give feedback before development began; the wireframes below reproduce that same low-fidelity, greyscale layout convention (boxes, placeholder text and control outlines rather than final visual styling).



The wireframe shows a dark header bar with 'DiseaseWatch KE' on the left and a 'Login' link on the right. Below the header is a light-colored box containing the 'Sign In' section. The 'Sign In' section has the subtitle 'Malaria & Cholera Outbreak Surveillance'. It includes a 'Username' label, a text input field with placeholder text 'Enter your username', a 'Password' label, a text input field with placeholder text 'Enter your password', and a dark 'LOG IN' button. Below the button is a link 'Don't have an account? Register here'. At the bottom of the box, it says 'Demo access: demo / demo123'.

Figure 3-15: Login screen wireframe

Create Account

Full Name

Jane Wanjiru

Username

jwanjiru

Email Address

jane@health.go.ke

Password

Confirm Password

County

-- Select County --

Organization

Ministry of Health

CREATE ACCOUNT

Already have an account? [Log in](#)

Validation: required fields, password match, min. 6 characters, unique username / email

Figure 3-16: Registration screen wireframe

DiseaseWatch KE

Dashboard

Dashboard

Predictions

Reports

Settings

Logout

National Malaria Risk

42.3%

▲ +2.1%

National Cholera Risk

31.8%

▼ -0.8%

Active Hotspots

6

▲ +1

Last Update

10:24 AM

Disease Hotspot Map — Kenya (47 Counties)

[Leaflet Choropleth Map]

County Analysis Controls

County

-- Select County --

Temperature (°C)

26°C

Rainfall (mm)

100 mm

Humidity (%)

65%

Population Density (/km²)

500

Sanitation Score (%)

65%

RUN PREDICTION

Prediction Results

Malaria Risk Assessment

72.4%

Level: HIGH

Cholera Risk Assessment

38.1%

Level: MEDIUM

7-Day Risk Forecast & Recommendations

Figure 3-17: Main dashboard wireframe — hotspot map, environmental parameter controls and prediction results

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DiseaseWatch KE

Predictions

Dashboard
Predictions
Reports
Settings
Logout

Prediction Input Form

County
-- Select County --

Temperature (°C)
26°C

Rainfall (mm)
100 mm

Humidity (%)
65%

Population Density (/km²)
500 /km²

Sanitation Score (%)
65%

RUN PREDICTION

Auto-fills temperature / rainfall / humidity from county weather API on selection.

Risk Assessment Results — Output

Malaria Risk
72.4%
Level: HIGH

Cholera Risk
38.1%
Level: MEDIUM

Environmental Impact Analysis

Breeding Potential68%

Temp SuitabilityHigh

Rainfall IndexModerate

Actionable Recommendations

7-Day Forecast

Figure 3-18: Predictions page wireframe — input form (left) and risk output panel (right)

DiseaseWatch KE

Reports

Dashboard
Predictions
Reports
Settings
Logout

Generate Report — Input Form

County
-- All Counties --

Report Type
-- Monthly Summary --

Month / Period
July 2026
GENERATE

Generated Reports — Output List

Nairobi — Monthly Summary
Period: July 2026 | Format: PDF
VIEW

Kisumu — Outbreak Risk Report
Period: June 2026 | Format: PDF
VIEW

Mombasa — Cholera WASH Report
Period: June 2026 | Format: PDF
VIEW

National — Quarterly Trend
Period: Q2 2026 | Format: PDF
VIEW

Figure 3-19: Reports page wireframe — report filter form and generated-report output list

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DiseaseWatch KE
Settings

Dashboard
Predictions
Reports
Settings
Logout

Profile
Notifications
Appearance
Security

Profile Information

Full Name

Username (read-only)

Email Address

Organization

County

SAVE PROFILE

Alert Notification Settings

High Risk Threshold

☒ Email
☒ SMS alerts (Phone Number field)

Alert Email Address

Security

Current Password

New Password

Confirm New Password

UPDATE PASSWORD

Appearance

Theme

Default Map Style

☒ Show hotspots
☒ Show risk le

Figure 3-20: Settings page wireframe — profile, notification, security and appearance panels

Design rationale: sliders were chosen over free-text numeric fields for environmental parameters to constrain input to realistic ranges and reduce user error; the county dropdown auto-populates default weather values through the weather interface so that officers without live sensor data can still run an estimate; risk outputs use a consistent red/blue colour convention (malaria/cholera) with a high-contrast score, level label and narrative description so that a health officer scanning the screen can grasp the outbreak risk within seconds.

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APPENDICES

Appendix I: Project Resources

Hardware Requirements for the Project:

- i. High-performance workstation/laptop.
- ii. Cloud server (virtual machine).
- iii. External storage devices (HDD/SSD).
- iv. Uninterruptible power supply (UPS).
- v. Networking equipment (router/modem).
- vi. GPU-enabled machine (for machine learning training).

Software Resource Requirements for the Project:

- i. Python – for data processing and machine learning development.
- ii. Pandas – for data cleaning and manipulation.
- iii. NumPy – for numerical computations.
- iv. TensorFlow – for building and training ML models.
- v. Keras – for designing neural networks (LSTM).
- vi. Flask or FastAPI – for backend API development.
- vii. React – for frontend user interface Leaflet – for GIS visualization.
- viii. SQLite(with PostGIS) – for data storage.
- ix. Git and GitHub – for version control and collaboration.

Appendix II: Project Budget

Table 0-1: Project Budget

Item	Description	Quantity	Unit Cost (Ksh)	Total Cost (Ksh)
Laptop	Development machine	1	60,000	60,000
Internet	Monthly subscription	3 months	2,000	6,000
Software	Licenses	-	0	0
Printing	Documentation	-	2,000	2,000
Miscellaneous	Contingency	-	2,000	2,000
Research costs	Research	3 months	2,000	6,000

Total Estimated Cost = Ksh. 76,000

Appendix III: Project Schedule

DURATION	Jan-26				Feb-26				Mar-26				Apr-26				May-26				Jun-26				Jul-26				Aug-26			
	Weeks				Weeks				Weeks				Weeks				Weeks				Weeks				Weeks							
ACTIVITIES	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
Requirements Gathering (Business Understanding)																																
Project Proposal																																
Proposal Defense																																
Data Understanding																																
Data Preparation																																
System & Database Design (Architecture, DFD, UML)																																
Modeling (ML Training)																																
Evaluation (Model & System Testing)																																
UI/UX Design & Implementation																																
Deployment & Documentation																																
Project Presentation																																

Figure 0-1: Project Gantt Chart

APPENDIX IV

Questionnaire for Requirements Elicitation Machine Learning-Powered Disease Surveillance System for Predicting Malaria and Cholera Outbreaks in Kenyan Counties

This questionnaire is intended for public health professionals involved in disease surveillance and health data management. Please answer all questions honestly. Your responses are strictly confidential and will be used solely for academic research purposes. Do not write your name on this form.

*** Indicates a required question**

Section A: Respondent Background Information

1. What is your professional designation? *

- ☐ Health Information Officer
- ☐ Public Health Nurse
- ☐ Disease Surveillance Coordinator
- ☐ Other: _____

2. How many years of experience do you have in public health or disease surveillance? *

- ☐ Less than 2 years
- ☐ 2 – 5 years
- ☐ 6 – 10 years
- ☐ More than 10 years

3. What is your highest level of education? *

- ☐ Certificate
- ☐ Diploma

- Bachelor's Degree
- Master's Degree or above

4. At what level do you primarily work? *

- Health Facility Level
- Sub-County Level
- County Level
- National Level

Section B: Current Surveillance Systems

Please indicate your level of agreement with the following statements. Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree

5. The current surveillance systems (e.g., DHIS2, HMIS) provide timely outbreak information. *

<i>Strongly Disagree</i>	1	2	3	4	5	<i>Strongly Agree</i>
	○	○	○	○	○	

6. Existing systems adequately support early detection of malaria and cholera outbreaks. *

<i>Strongly Disagree</i>	1	2	3	4	5	<i>Strongly Agree</i>
	○	○	○	○	○	

7. Data reported through current systems is complete and consistently formatted. *

<i>Strongly Disagree</i>	1	2	3	4	5	<i>Strongly Agree</i>
	○	○	○	○	○	

8. Current systems integrate data from multiple sources such as climate and environmental data. *

<i>Strongly Disagree</i>	1	2	3	4	5	<i>Strongly Agree</i>
	○	○	○	○	○	

9. Existing dashboards and reports are easy to interpret and act upon. *

<i>Strongly Disagree</i>	1	2	3	4	5	<i>Strongly Agree</i>
	○	○	○	○	○	

Section C: Limitations of Existing Systems

10. Which of the following limitations have you experienced with existing surveillance systems? *

(Tick all that apply)

- ☐ Delayed reporting and data lag
- ☐ Lack of real-time data integration
- ☐ No predictive or early warning capability
- ☐ Poor geospatial visualization
- ☐ Underreporting from health facilities
- ☐ Other: _____

Section D: Proposed System Features

Please rate the importance of the following proposed features for the new surveillance system. Scale: 1 = Not Important, 2 = Slightly Important, 3 = Moderately Important, 4 = Important, 5 = Very Important

11. County-level outbreak risk scores updated regularly. *

<i>Not Important</i>	1	2	3	4	5	<i>Very Important</i>
	☐	☐	☐	☐	☐	

12. Interactive geospatial maps showing outbreak risk by county. *

<i>Not Important</i>	1	2	3	4	5	<i>Very Important</i>
	☐	☐	☐	☐	☐	

13. Automated alert notifications when outbreak risk exceeds a threshold. *

<i>Not Important</i>	1	2	3	4	5	<i>Very Important</i>
	☐	☐	☐	☐	☐	

14. Integration with existing DHIS2 data pipelines. *

<i>Not Important</i>	1	2	3	4	5	<i>Very Important</i>
	☐	☐	☐	☐	☐	

15. Historical trend visualization for malaria and cholera cases. *

<i>Not Important</i>	1	2	3	4	5	<i>Very Important</i>
	☐	☐	☐	☐	☐	

16. A mobile-accessible dashboard for use in the field. *

Not Important	1	2	3	4	5	Very Important
	☐	☐	☐	☐	☐	

17. Role-based access control to restrict sensitive data by user role. *

Not Important	1	2	3	4	5	Very Important
	☐	☐	☐	☐	☐	

Section E: System Usability and Adoption

Please indicate your level of agreement with the following statements. Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree

18. I would be comfortable using a web-based dashboard for outbreak surveillance. *

Strongly Disagree	1	2	3	4	5	Strongly Agree
	☐	☐	☐	☐	☐	

19. I would trust outbreak predictions generated by a machine learning model if they were validated and explained clearly. *

Strongly Disagree	1	2	3	4	5	Strongly Agree
	☐	☐	☐	☐	☐	

20. A predictive surveillance system would improve my department's ability to respond to outbreaks proactively. *

Strongly Disagree	1	2	3	4	5	Strongly Agree
	☐	☐	☐	☐	☐	

21. My department has adequate internet connectivity to support a web-based surveillance platform. *

Strongly Disagree	1	2	3	4	5	Strongly Agree
	☐	☐	☐	☐	☐	

22. Training and onboarding for a new digital health platform would be manageable within my department's capacity. *

Strongly Disagree	1	2	3	4	5	Strongly Agree
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Section F: Open Ended Questions

23. In your opinion, what is the single most important feature a disease outbreak prediction system should have? *

24. What concerns, if any, do you have about the use of machine learning for disease outbreak prediction in Kenya? *

25. Are there any additional comments or suggestions you would like to share regarding the proposed surveillance system?

Thank you for taking the time to complete this questionnaire.